

Discussion paper for CIE review of the eastern Bering Sea walleye pollock stock assessment

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Executive Summary

The work presented below was based on the same data sets used in the 2024 assessment. The first task was to update the model runs to include the 2023 data. The model runs were updated and the results are presented below. The model runs were then used to compare the FT-NIR data with the traditional data sources (BTS and ATS). The results of these comparisons are presented below. The model fits to the data are presented in the tables at the end of the document. The first section steers towards the R library used for processing model results for EBS pollock. The second section presents the comparisons of the FT-NIR data with the traditional data sources. The third section compares results of different software platforms (e.g., SS3, CEA, and AMAK) for the assessment. The fourth section shows results comparing newer model-based estimates from survey data.

R Library for assessment compilations (ebswp)

The **ebswp** library is a collection of functions for compiling and analyzing results from the stock assessment models for the eastern Bering Sea walleye pollock (Ianelli (2025)). The library is designed to work with the ADMB code and depends on a consistent format for the report files generated. The link in the reference provides access to some notes on applications as vignettes to the package (e.g., . [here](#)).

Comparing FT-NIR compositions with traditional microscope age estimations

The fish age and growth lab at the AFSC have endeavored to expand on new technologies (Helser et al. (2019)).

FT-NIRS data consistency comparisons

Examining the abundance-at-age (for design-based estimates of the bottom-trawl age compositions) we show that in general, the TMA estimates show a high level of consistency by cohort (Figure 1). For these same data, when applied to the FTNIRS age-estimation method, the consistency was lower (Figure 2). The extent that this is a concern is unknown.

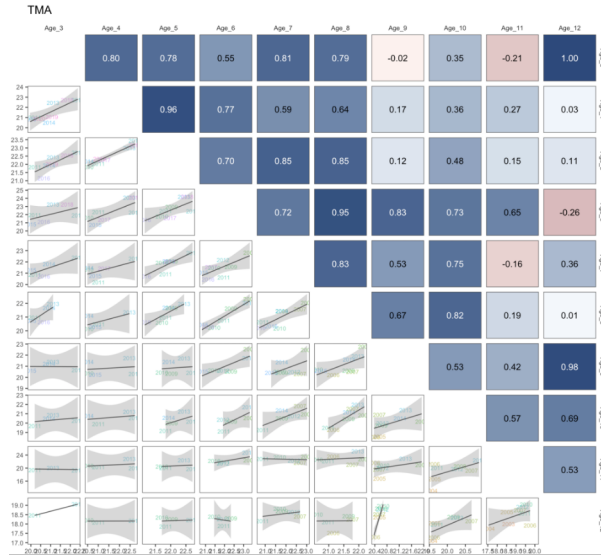


Figure 1: Cohort consistency over different ages based on log-abundance from the bottom-trawl survey data for pollock in recent years for traditional microscope age estimates. The cohorts are shown in the labels.

When preparing the data for the assessment, we plotted out the proportions-at-age for the TMA data and noted the differences in the estimates using the FTNIRS age estimates (Figure 3,

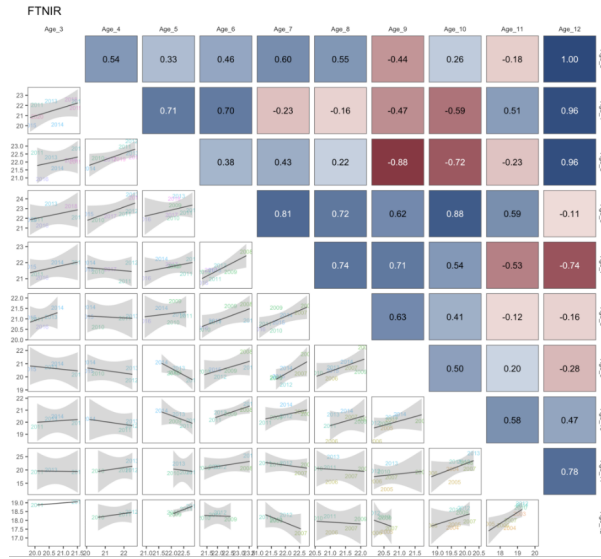


Figure 2: Cohort consistency over different ages based on log-abundance from the bottom-trawl survey data for pollock in recent years for FTNIRs age estimates. The cohorts are shown in the labels.

Figure 4, and Figure 5). Other preparations involved estimating the age-determination errors for each method following that of Punt et al. (2008). For the FTNIRs data, we optionally estimated separate age-estimation error matrices by gear type. The appearance of the fleet-aggregated and fleet-specific input data files is shown in Figure 6.

For the assessment model, the alternatives are shown based on the coverage of data and availability among types of ageing is shown in Figure 7. These figures are intended to show what data are available for the model testing and the different ways of treating age determination error matrices.

Comparisons of fit to FT-NIRS models

The layout of model runs comparing the traditional microscope ages (TMA) with the FT-NIRS ages is shown in Table 1. The models are run with the FT-NIRS data with different age-error matrix specifications. The model fits to the data are shown in Table 2. The table shows the goodness-of-fit

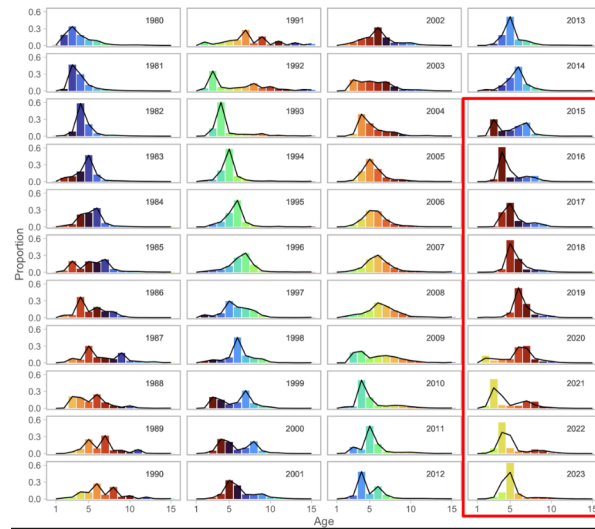


Figure 3: Proportions at age for recent years for the fishery data where the columns represent the traditional microscope age estimates (TMA) compared to the FTNIRs age estimates (lines for the red-outlined panels).

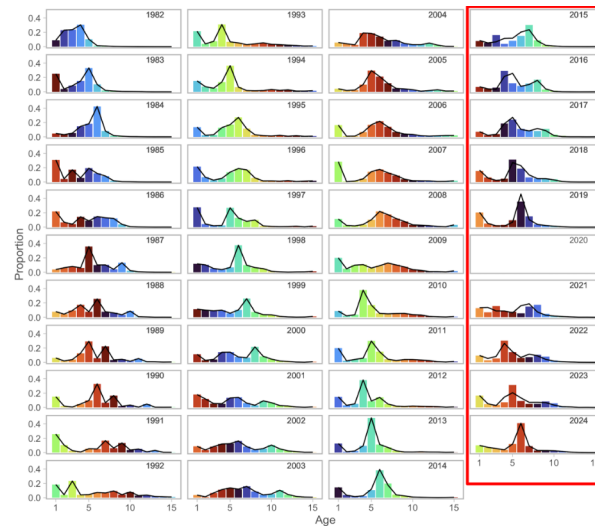


Figure 4: Proportions at age for recent years for the bottom-trawl survey data where the columns represent the traditional microscope age estimates (TMA) compared to the FTNIRs age estimates (lines for the red-outlined panels).

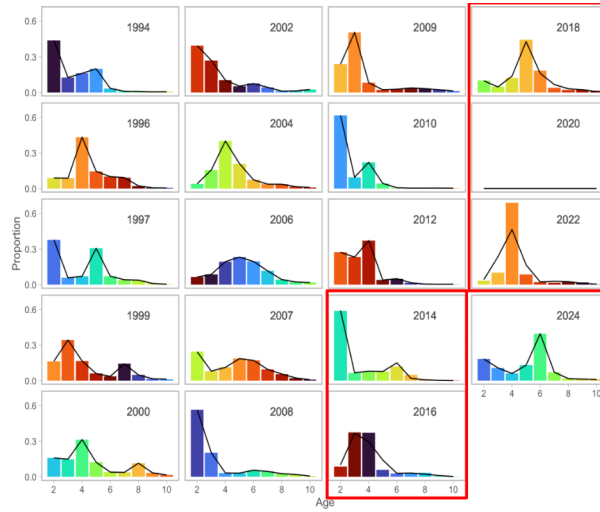


Figure 5: Proportions at age for recent years for the acoustic-trawl survey data where the columns represent the traditional microscope age estimates (TMA) compared to the FTNIRs age estimates (lines for the red-outlined panels).

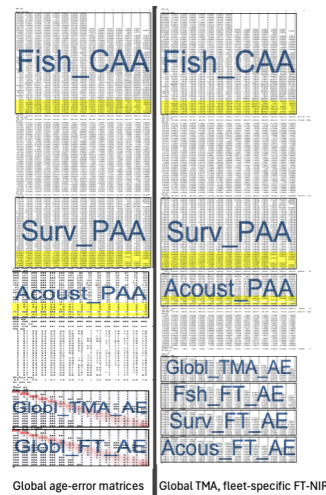


Figure 6: Schematic of input datafile showing the breakout of the fleet-specific FTNIRs age-error matrices (right panel) compared to that where the global FTNIRs age-error matrices are used (left panel).

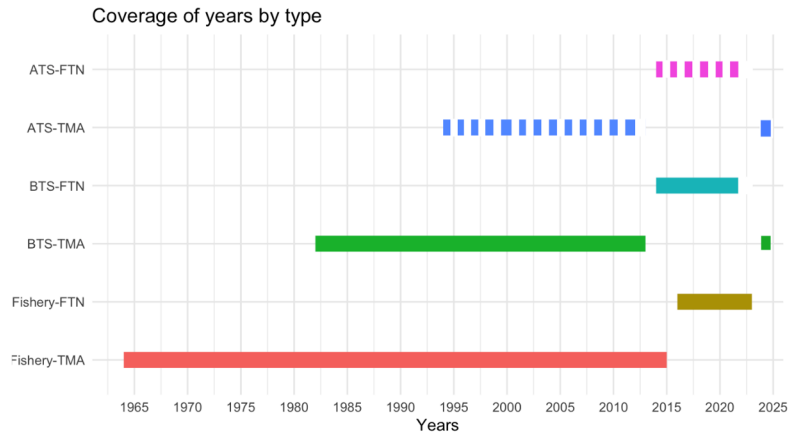


Figure 7: Data extent by age composition for the models with FTNIRs age estimates. The data types are for the Fishery, bottom trawl survey (BTS), and acoustic trawl survey (ATS) and the years of coverage are shown in the x-axis.

Table 1: Configuration of different model options testing the impact of different treatment of the age composition data and ageing error.

Name	Model description	
	Age type	Age-error
Base	TMA	Global TMA
FT-NIR 1	TMA + FT-NIR	Global TMA, global FT-NIR
FT-NIR 2	TMA + FT-NIR	Global TMA, disaggregated FT-NIR

Table 2: Goodness-of-fit measures to primary data for different data and age-error applications. RMSE=root-mean square log errors, NLL=negative log-likelihood, SDNR=standard deviation of normalized residuals, Eff. N=effective sample size for composition data). See text for incremental model descriptions.

Component	Base	FT-NIR fleets aggregated	FT-NIR fleet-specific
RMSE BTS	0.150	0.151	0.152
RMSE ATS	0.167	0.163	0.158
RMSE AVO	0.216	0.217	0.215
RMSE CPUE	0.097	0.097	0.097
SDNR BTS	0.890	0.900	0.880
SDNR ATS	0.930	0.910	0.880
SDNR AVO	0.950	0.960	0.940
Eff. N Fishery	358	336	381
Eff. N BTS	159	148	153
Eff. N ATS	233	221	233
Catch NLL	3	4	3
BTS NLL	24	24	23
ATS NLL	8	7	7
ATS Age1 NLL	7	8	9
AVO NLL	8	8	8
Fish Age NLL	417	559	614
BTS Age NLL	286	330	313
ATS Age NLL	43	42	41
NLL selectivity	213	216	215
NLL Priors	20	20	20
Data NLL	807	995	1,032
Total NLL	1110.900	1298.930	1332.810

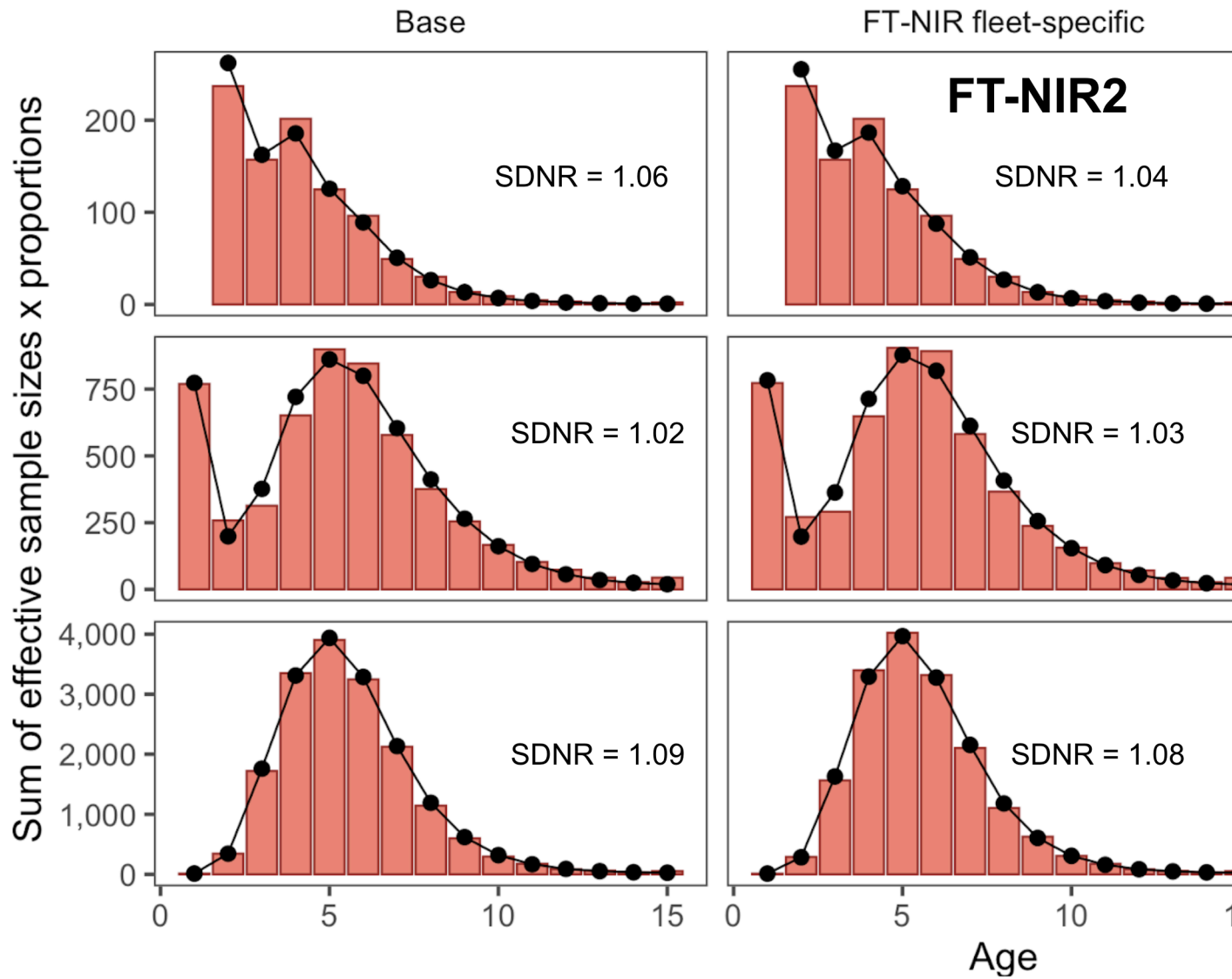


Figure 8: One-step ahead residuals for different model specifications for the fishery composition data

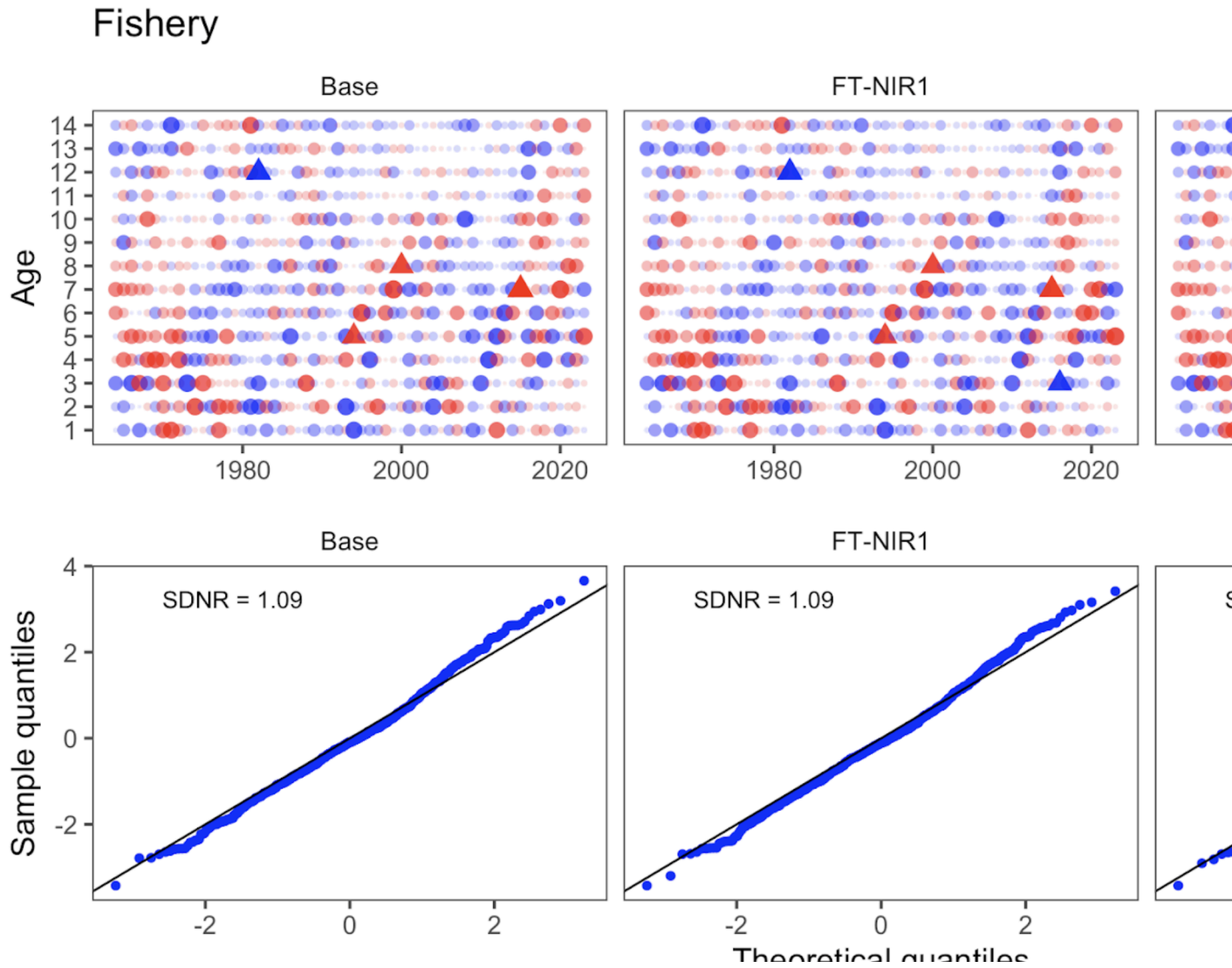


Figure 9: One-step ahead residuals for different model specifications for the fishery composition data

Bottom trawl survey

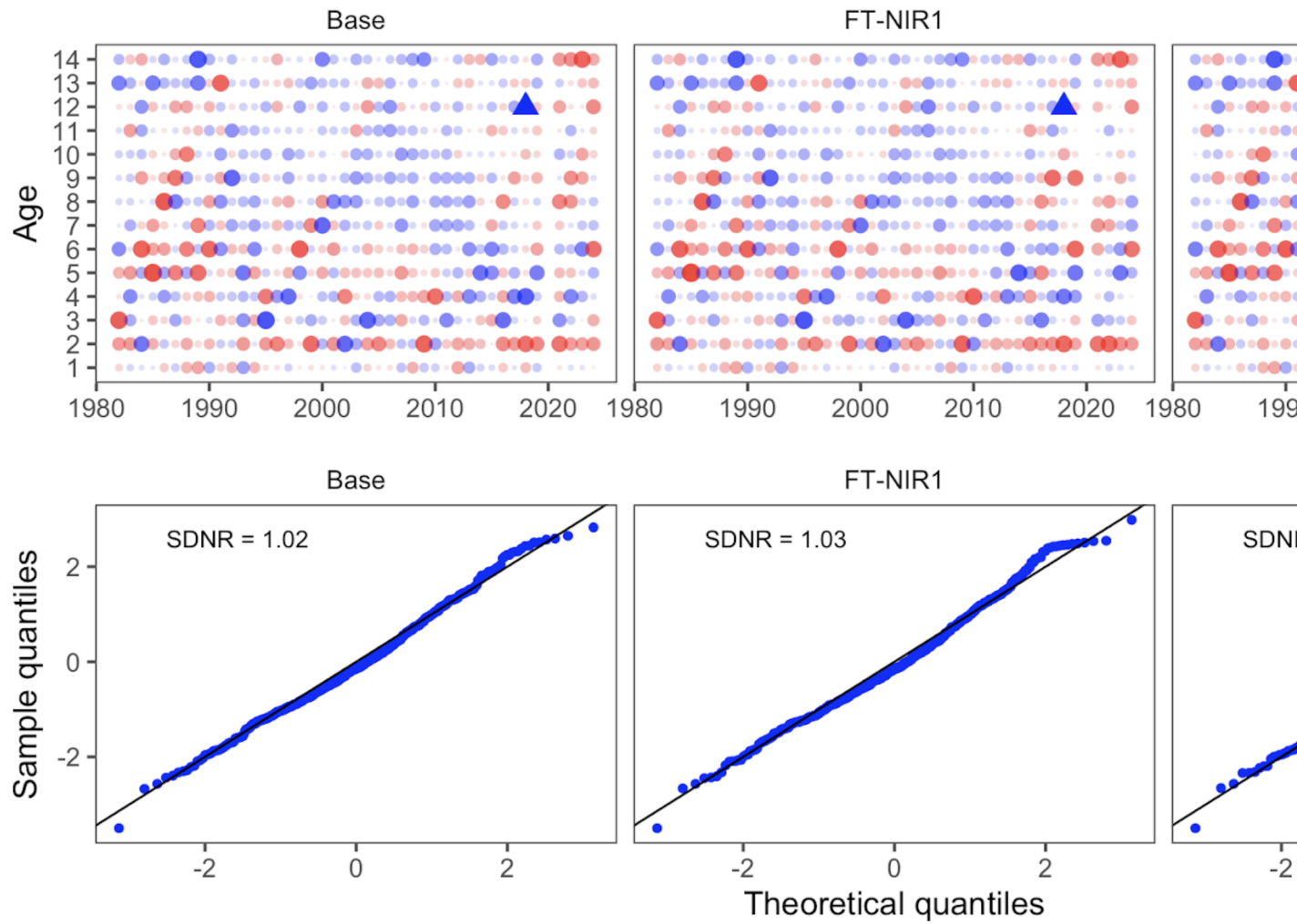


Figure 10: One-step ahead residuals for different model specifications for the bottom-trawl survey composition data

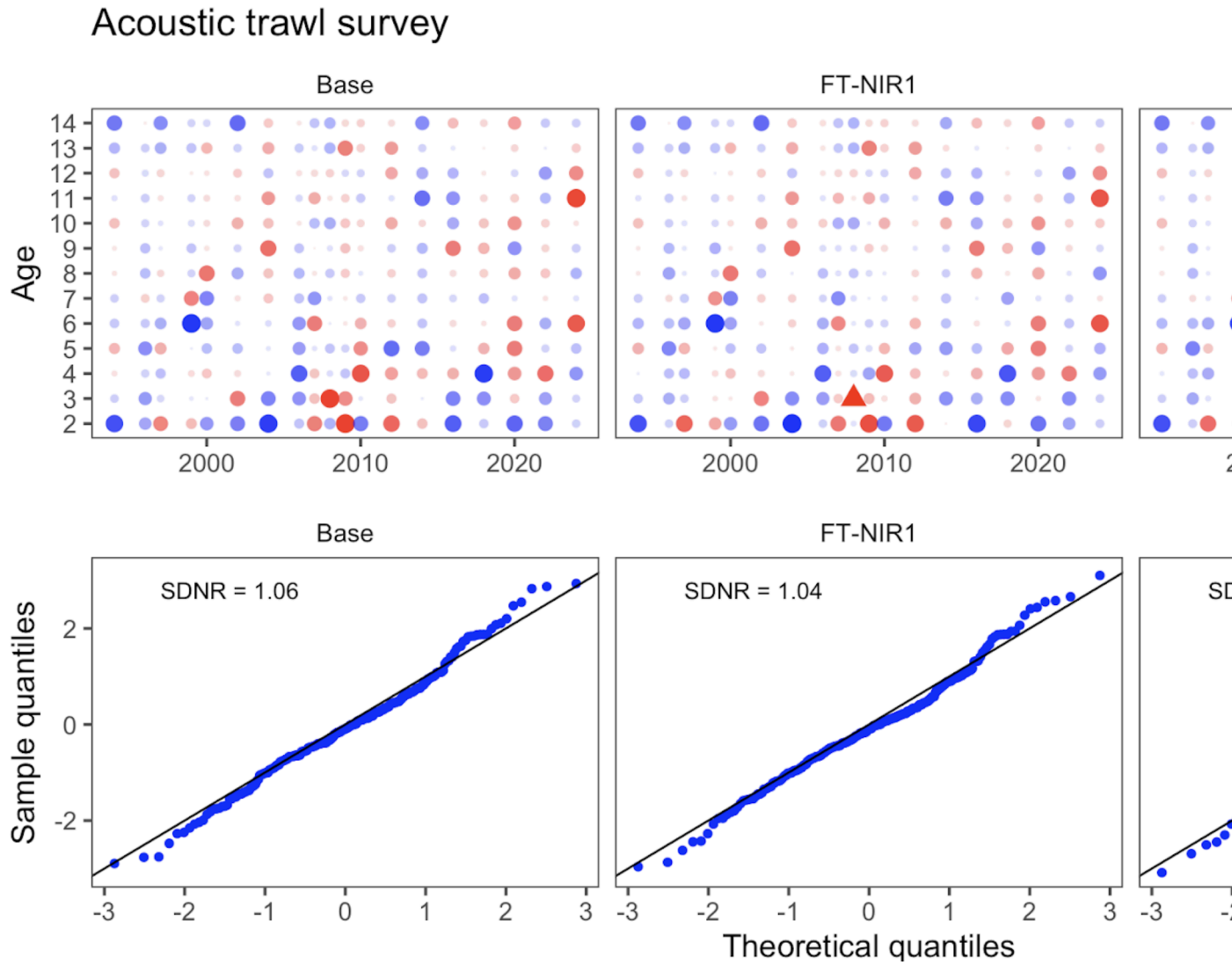


Figure 11: One-step ahead residuals for different model specifications for the acoustic-trawl survey composition data

Table 3: Goodness-of-fit measures to primary data for different data and age-error applications. RMSE=root-mean square log errors, NLL=negative log-likelihood, SDNR=standard deviation of normalized residuals, Eff. N=effective sample size for composition data). See text for incremental model descriptions.

Component	Base	FT-NIR fleets aggregated	FT-NIR fleet-specific
B_{2025}	2,800	2,500	2,700
$CV_{B_{2025}}$	0.14	0.14	0.14
B_{MSY}	1,667	1,652	1,670
$CV_{B_{MSY}}$	0.23	0.24	0.23
B_{2025}/B_{MSY}	165%	153%	162%
B_0	4,318	4,360	4,436
$B_{35\%}$	1,935	1,898	1,931
SPR rate at F_{MSY}	31%	30%	29%
Steepness	0.63	0.64	0.64
Est. $B_{2024}/B_{2024, \text{nofishing}}$	0.65	0.62	0.64
B_{2024}/B_{MSY}	182%	169%	178%

Leave-one-out survey sensitivity analysis

Here we present model runs that downweight one or more sources of survey data at a time. Table 1 shows the combinations of data types included for this analysis, which includes six model runs: (1) downweight all Acoustic Trawl Survey (ATS) data (age 1 survey index, age 2+ survey biomass index, and age composition data), (2) downweight the ATS age composition data, (3) downweight the Acoustic Vessels of Opportunity (AVO) data (survey biomass index), (4) downweight all Eastern Bering Sea Trawl Survey (BTS) data (age 1+ survey biomass index and age composition data), (5) downweight the BTS age composition data, and (6) downweight all survey biomass indices (BTS, ATS, and AVO survey biomass indices). The ATS age 1 data are included as a separate survey index in the assessment model, while BTS age 1 data are included with the age composition data in the multinomial objective function component.

Among these models, downweighting the BTS data (both solely downweighting the BTS age composition data and downweighting all BTS data) led to changes in trends over time in estimated spawning biomass Figure 12 . The model estimates of recruitment were variable across sensitivity runs, but estimates differed the most from the 2024 accepted model when all BTS data were downweighted Figure 13. Estimates of BTS survey biomass fit the BTS data very poorly in most years when the BTS data were downweighted (Figure 14). Additionally, the fits to the BTS survey biomass were affected by downweighting the BTS age composition data (without downweighting the BTS survey index). The model was relatively insensitive to other data being excluded, and fits to the ATS and AVO indices were relatively insensitive to downweighting the BTS data (see Figure 14, Figure 15, and Figure 16). The root mean square error (RMSE) in Table 2 for the model run that downweighted all BTS data was large and the effective sample size (Eff. N) was low as compared to the 2024 accepted model.

Table 1: Configuration of different model options testing the impact of different data components

Dataset	Model description					
	Drop all BTS	Drop all ATS	Drop all indices	Drop AVO	Drop BTS Age composition	Drop ATS Age composition
BTS age 2+ Index	X		X			
ATS age 2+ Index		X	X			
ATS age 1 Index		X	X			
AVO Index			X			
BTS comps	X				X	
ATS comps		X				

Additionally, downweighting solely the BTS age composition data led to low effective sample size for the BTS.

The importance of the survey data can be seen in some of the reference calculations where the uncertainty increases dramatically and the stock-status perception was quite different across sensitivity runs Table 3. Most dramatically, dropping all survey indices led to a CV in “ B_{MSY} ” estimates of 1.26 as compared to 0.13 in the 2024 accepted model.

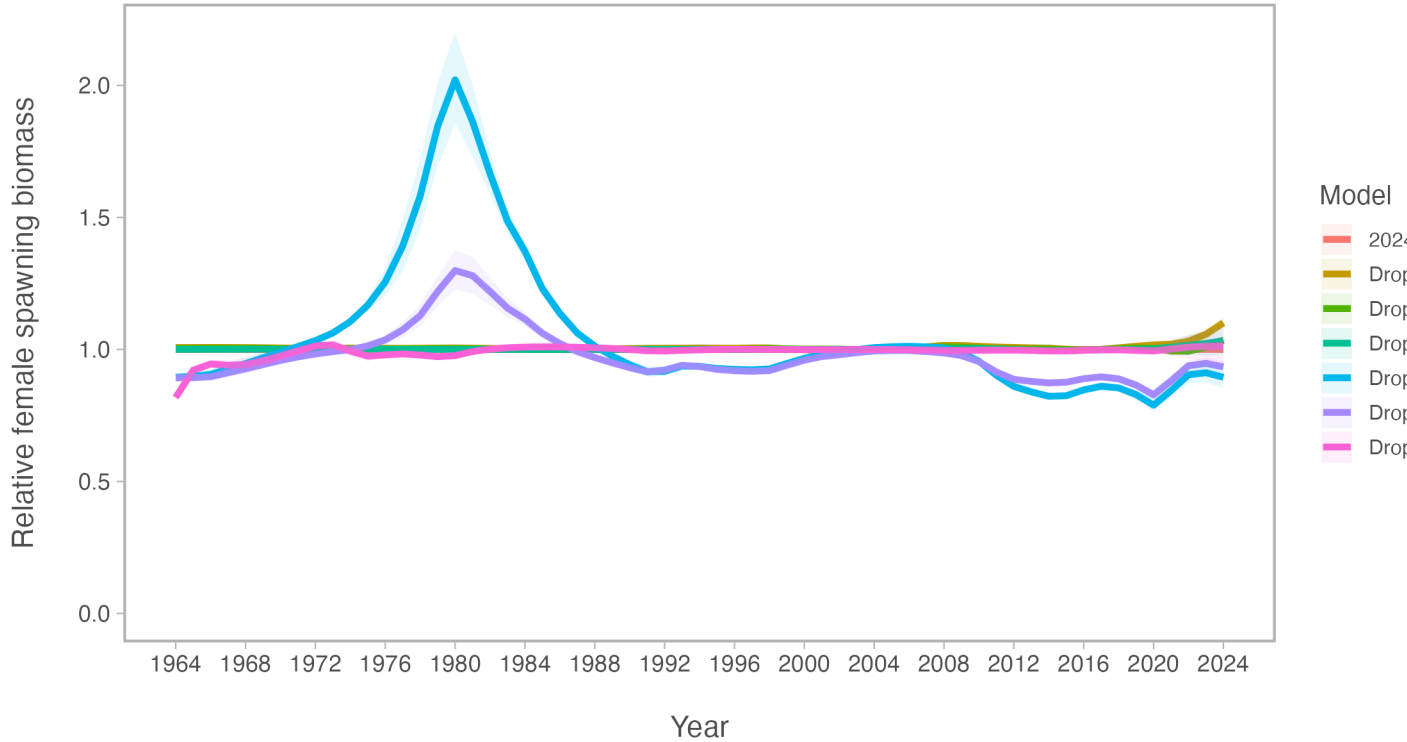


Figure 12: Comparison of spawning biomass (SSB) relative to the 2024 accepted model for models that downweight one data source at a time

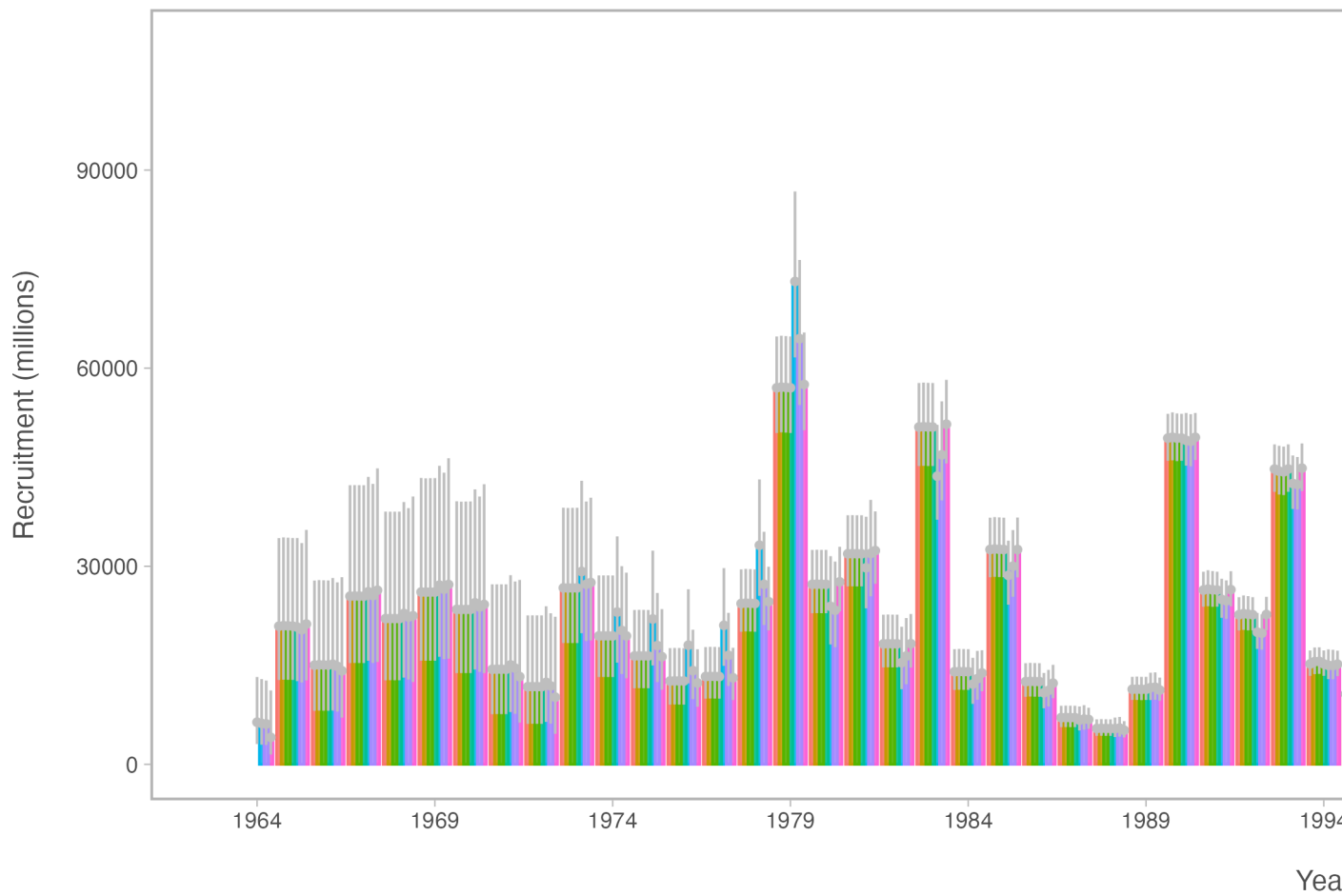


Figure 13: Comparison of recruitment estimates for models that downweight one data source at a time

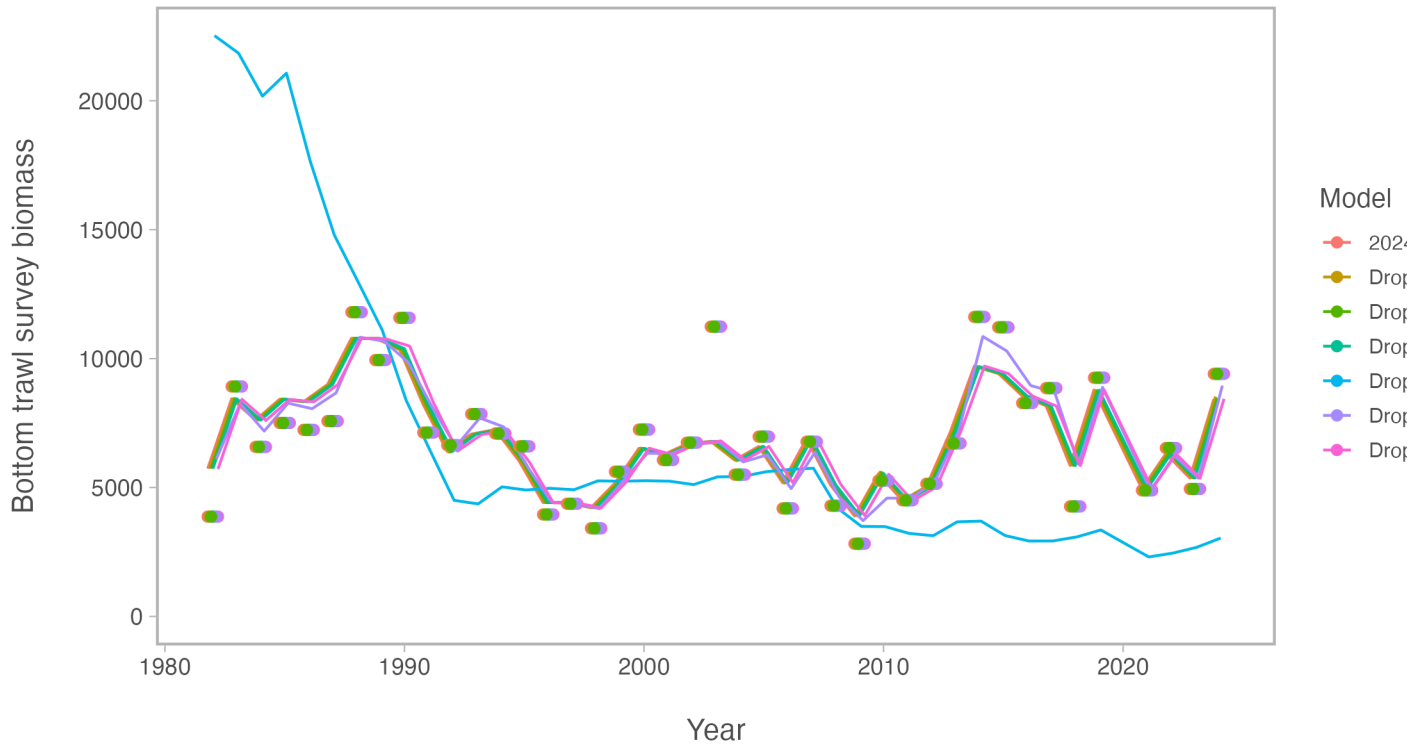


Figure 14: Comparison of fits to the Eastern Bering Sea Bottom Trawl Survey (BTS) biomass index for models that downweight one data source at a time

Table 2: Goodness-of-fit measures to primary data for leave-one-out model sensitivity runs. RMSE=root-mean square log errors, Eff. N=effective sample size for composition data).

Component	2024 accepted	Drop BTS	Drop ATS	Drop AVO	Drop indices	Drop BTS age
RMSE BTS	0.157	0.689	0.157	0.157	0.157	0.152
RMSE ATS	0.180	0.182	0.188	0.191	0.187	0.178
RMSE AVO	0.229	0.237	0.238	0.241	0.238	0.231
RMSE CPUE	0.093	0.090	0.093	0.093	0.098	0.092
Eff. N Fishery	1,182	1,379	1,190	1,185	1,176	1,344
Eff. N BTS	248	95	254	249	245	84
Eff. N ATS	269	230	164	269	273	228

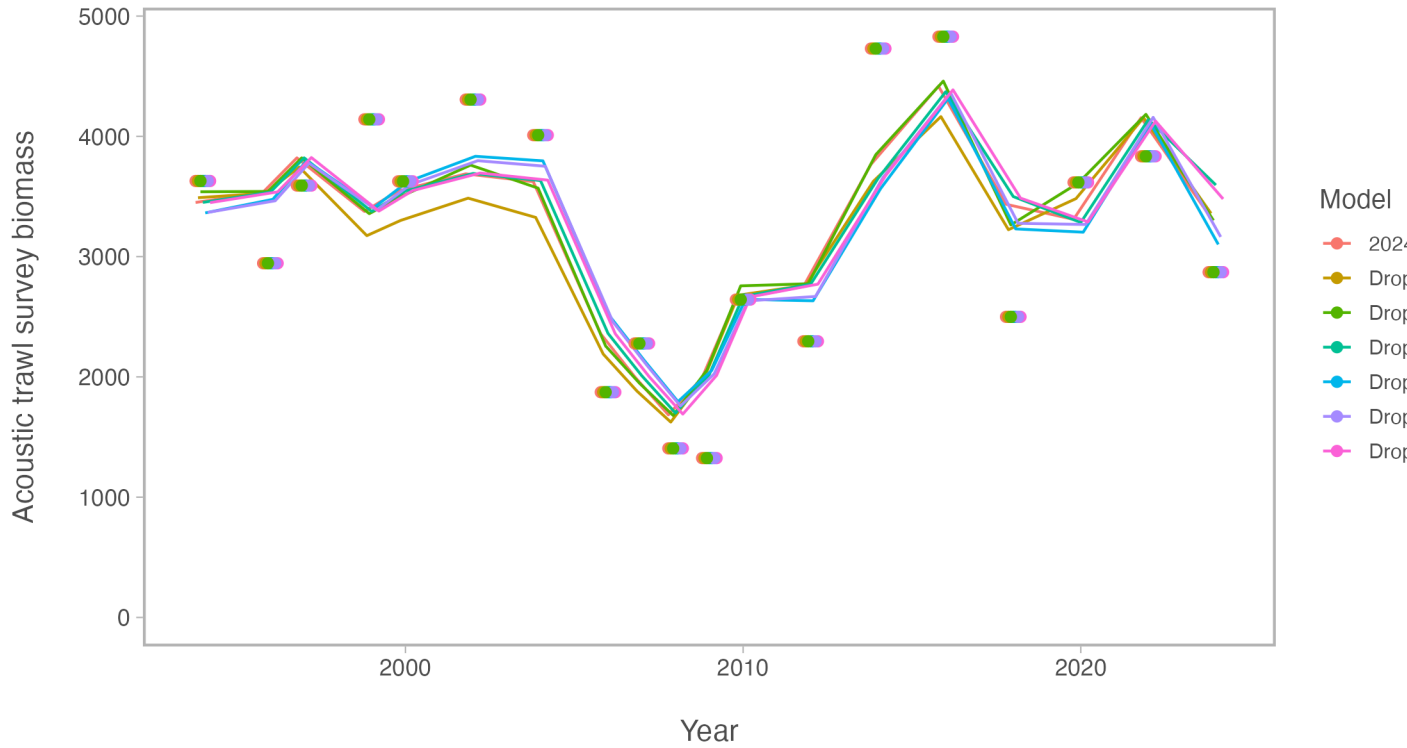


Figure 15: Comparison of fits to the Acoustic Trawl Survey (ATS) biomass index for models that downweight one data source at a time

Table 3: Estimated derived quantities and selected corresponding CVs for leave-one-out sensitivity analysis runs. See text for incremental model descriptions.

Estimated Quantity	2024 accepted	Drop BTS	Drop ATS	Drop AVO	Drop indices	Drop
B_{2025}	3,100	2,700	3,600	3,300	3,200	
$CV_{B_{2025}}$	0.13	0.16	0.14	0.16	0.18	
B_{MSY}	2,310	2,415	2,360	2,322	3,677	
$CV_{B_{MSY}}$	0.3	0.31	0.31	0.31	1.26	
B_{2025}/B_{MSY}	135%	111%	151%	141%	86%	
B_0	5,975	6,136	6,120	6,009	8,934	
$B_{35\%}$	2,066	1,988	2,101	2,080	2,074	
SPR rate at F_{MSY}	31%	35%	31%	31%	42%	
Steepness	0.62	0.59	0.62	0.62	0.52	
Est. $B_{2024}/B_{2024,nofishing}$	0.61	0.51	0.63	0.61	0.42	
B_{2024}/B_{MSY}	147%	126%	158%	151%	93%	

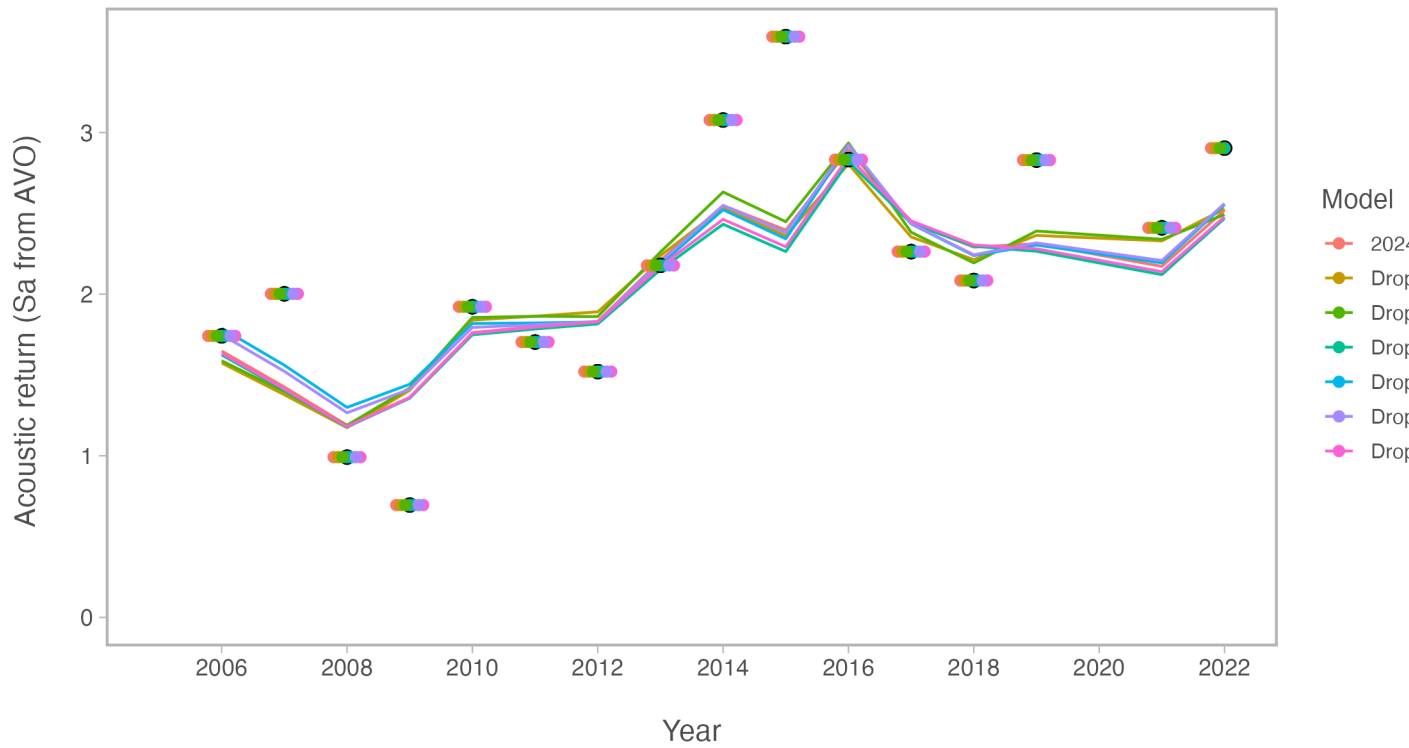


Figure 16: Comparison of fits to the Acoustic Vessels of Opportunity (AVO) biomass index for models that downweight one data source at a time

Incorporating multi-species M-at-age and year estimates, also total uncertainty in acoustics...

A commonly adopted approach is to use estimated natural mortality-at-age and year from multispecies trophic interaction models (e.g., Trijoulet, Fay, and Miller (2020)). The current assessment has an option to include the 2024 CEATTLE estimates of natural mortality (Table 1; Figure 17). This resulted in slightly higher recruitment (Figure 18) lower spawning biomass in the near term (Figure 19). This is due to the higher natural mortality for most ages and years compared to the base 2024 model.

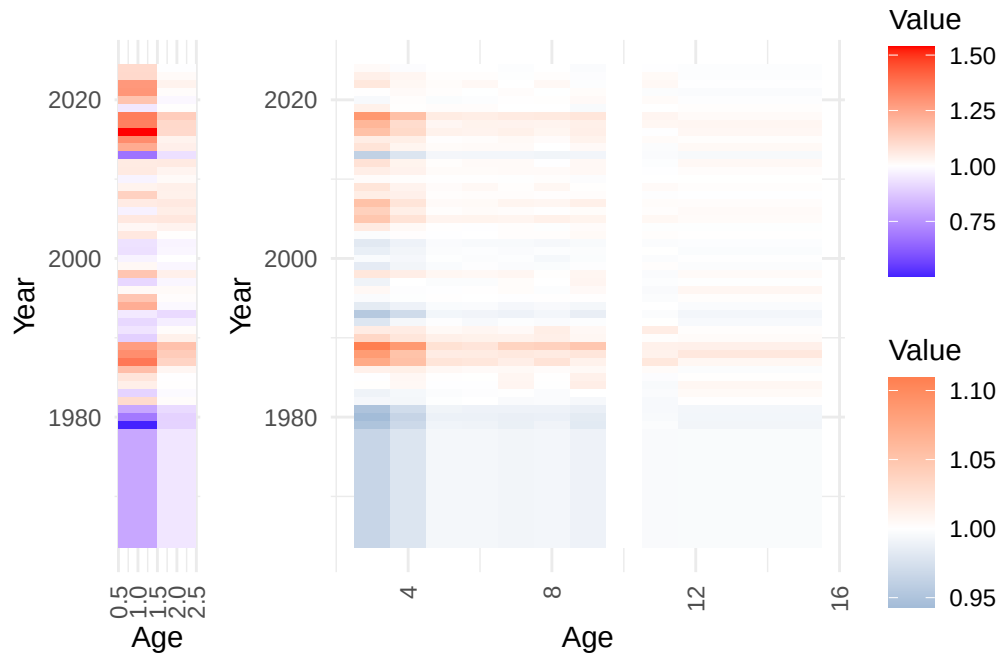


Figure 17: Depiction of the natural mortality (variability from mean value at age) as estimated from CEATTLE.

We also include a comparison of revised acoustic-trawl estimates (Urmy, Ressler, and De Robertis (2025); Figure 20). The natural mortality alternative run resulted in generally better fits to most data components whereas the “untuned” run with the Urmy, Ressler, and De Robertis (2025) estimates were generally poorer (Table 2). Results from the Urmy, Ressler, and De Robertis (2025) had minor impact on management-related quantities whereas those

Table 1: Natural mortality-at-age used over ages and time based on the CEATTLE 2024 model.

	1	2	3	4	5	6	7	8	9	10
1964	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1965	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1966	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1967	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1968	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1969	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1970	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1971	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1972	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1973	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1974	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1975	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1976	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1977	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1978	1.024	0.341	0.320	0.311	0.303	0.304	0.303	0.304	0.304	0.3001
1979	0.644	0.326	0.315	0.308	0.303	0.303	0.302	0.303	0.302	0.3001
1980	0.883	0.325	0.313	0.307	0.302	0.302	0.302	0.303	0.302	0.3001
1981	1.022	0.333	0.315	0.309	0.303	0.303	0.303	0.303	0.303	0.3001
1982	1.415	0.365	0.329	0.316	0.305	0.306	0.305	0.305	0.305	0.3001
1983	1.156	0.358	0.328	0.316	0.305	0.306	0.306	0.305	0.307	0.3001
1984	1.342	0.363	0.332	0.320	0.305	0.306	0.308	0.306	0.312	0.3001
1985	1.380	0.362	0.331	0.320	0.306	0.306	0.308	0.307	0.311	0.3001
1986	1.517	0.371	0.336	0.322	0.307	0.307	0.308	0.307	0.309	0.3001
1987	1.756	0.406	0.356	0.335	0.312	0.312	0.310	0.314	0.311	0.3001
1988	1.684	0.415	0.360	0.335	0.310	0.311	0.312	0.312	0.314	0.3001
1989	1.632	0.425	0.368	0.346	0.313	0.313	0.317	0.318	0.322	0.3001
1990	1.145	0.377	0.343	0.325	0.309	0.310	0.309	0.311	0.309	0.3001
1991	1.206	0.365	0.337	0.323	0.307	0.308	0.307	0.311	0.309	0.3001
1992	1.169	0.348	0.325	0.316	0.305	0.305	0.305	0.305	0.307	0.3001
1993	1.220	0.332	0.316	0.309	0.303	0.303	0.303	0.304	0.303	0.3001
1994	1.576	0.357	0.326	0.315	0.304	0.305	0.304	0.305	0.305	0.3001
1995	1.493	0.365	0.331	0.318	0.306	0.306	0.306	0.307	0.307	0.3001
1996	1.306	0.367	0.334	0.318	0.305	0.306	0.307	0.306	0.309	0.3001
1997	1.168	0.355	0.328	0.318	0.305	0.305	0.307	0.306	0.310	0.3001
1998	1.492	0.377	0.338	0.323	0.308	0.308	0.308	0.307	0.310	0.3001
1999	1.311	0.355	0.326	0.316	0.304	0.306	0.305	0.306	0.307	0.3001
2000	1.249	0.362	0.330	0.316	0.305	0.305	0.305	0.305	0.306	0.3001
2001	1.194	0.355	0.327	0.316	0.305	0.305	0.305	0.306	0.306	0.3001
2002	1.199	0.353	0.325	0.315	0.304	0.305	0.305	0.305	0.306	0.3001
2003	1.367	0.364	0.331	0.319	0.305	0.306	0.306	0.307	0.309	0.3001
2004	1.312	0.374	0.337	0.320	0.306	0.306	0.306	0.306	0.308	0.3001 ²²
2005	1.366	0.387	0.348	0.326	0.308	0.309	0.309	0.310	0.310	0.3001
2006	1.243	0.380	0.345	0.323	0.306	0.307	0.307	0.307	0.308	0.3001
2007	1.355	0.385	0.349	0.325	0.307	0.307	0.309	0.309	0.311	0.3001
2008	1.450	0.377	0.336	0.322	0.306	0.307	0.307	0.307	0.308	0.3001
2009	1.328	0.378	0.339	0.321	0.306	0.308	0.306	0.308	0.307	0.3001
2010	1.250	0.368	0.333	0.319	0.306	0.306	0.306	0.307	0.307	0.3001

EBS pollock assessment discussion paper

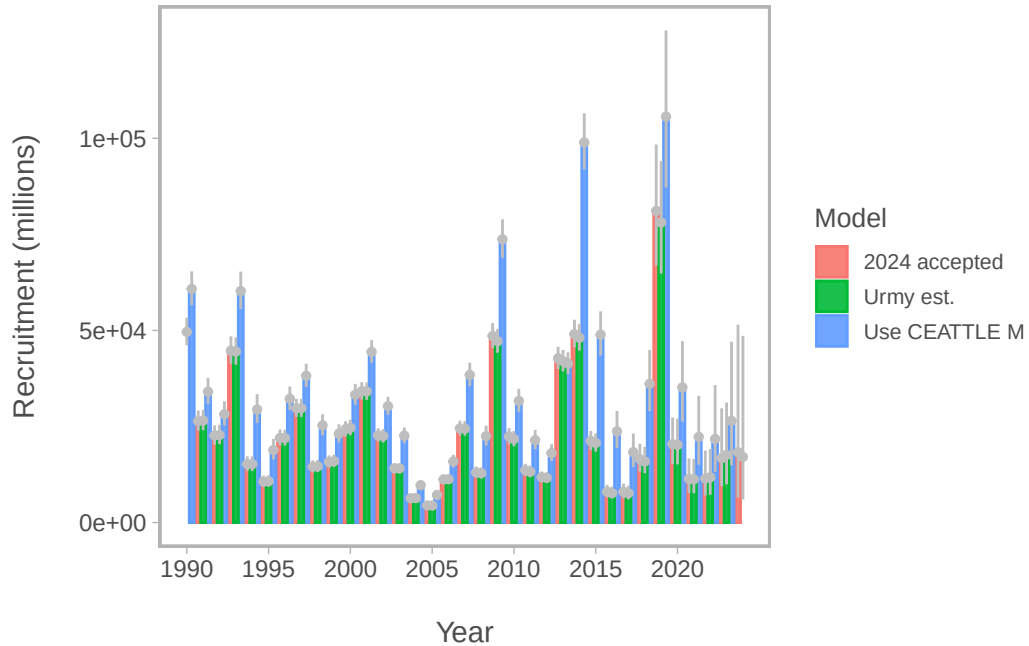


Figure 18: Comparison of recruitment estimates comparing 2024 assessment results with one where M-at-age and year is specified from CEATTLE.

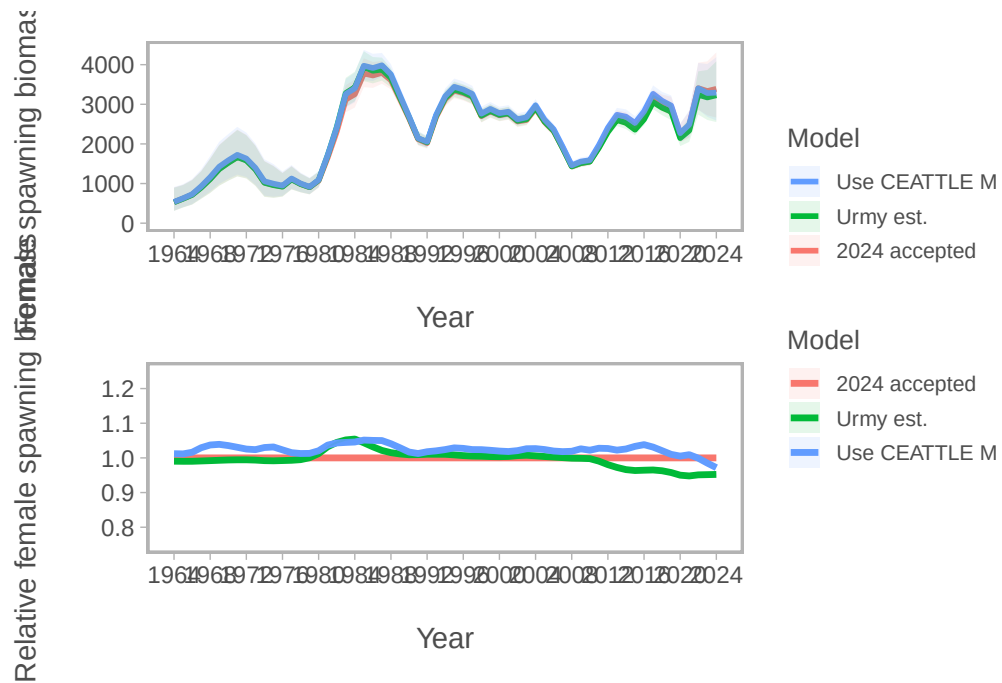


Figure 19: Comparison of spawning biomass (SSB) and SSB relative to the 2024 accepted model for model that uses the M-at-age matrix from CEATTLE.

with the alternative natural mortality specification differed in a number of ways (Table 3). The most notable difference was the estimated B_{MSY} and $B_{35\%}$ values.

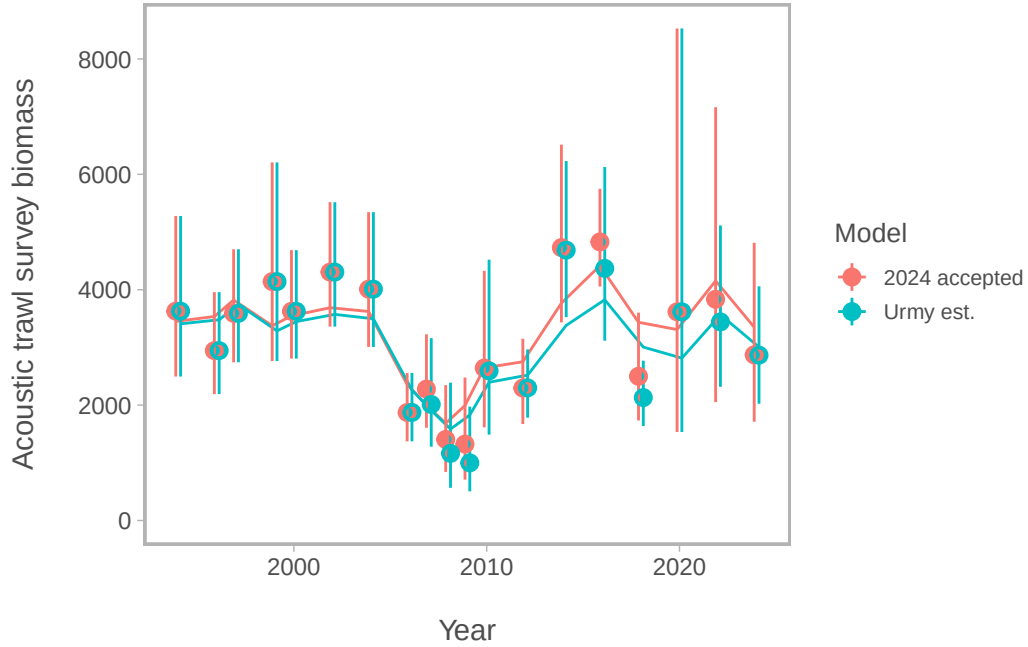


Figure 20: Comparison of fits to the Eastern Bering Sea acoustic-Trawl Survey (ATS) biomass index for the accepted 2024 model and one with revised estimates of index values and uncertainty from Urmy et al., (in prep).

Table 2: Goodness-of-fit measures to primary data for model sensitivity to applying the matrix of M-at-age and year and acoustic uncertainty estimates. RMSE=root-mean square log errors, SDNR=standard deviation of normalized residuals, Eff. N=effective sample size for composition data).

Component	2024 accepted	Use CEATTLE M	Urmy est.
RMSE BTS	0.157	0.157	0.161
RMSE ATS	0.180	0.179	0.226
RMSE AVO	0.229	0.230	0.236
RMSE CPUE	0.093	0.093	0.093
SDNR BTS	0.950	0.960	0.970
SDNR ATS	1.000	0.990	1.220
SDNR AVO	0.980	0.960	1.020
Eff. N Fishery	1,182	1,176	1,188
Eff. N BTS	248	241	221
Eff. N ATS	269	267	278
Catch NLL	3	3	3
BTS NLL	31	31	30
ATS NLL	9	9	13
ATS Age1 NLL	11	12	14
AVO NLL	9	8	9
Fish Age NLL	167	168	162
BTS Age NLL	194	196	205
ATS Age NLL	30	31	33
NLL selectivity	196	196	199
NLL Priors	20	20	20
Data NLL	465	468	480
Total NLL	718	713	735

Table 3: Estimated derived quantities and selected corresponding CVs for sensitivity to natural mortality specification and acoustic uncertainty estimates.

Estimated Quantity	2024 accepted	Use CEATTLE M	Urmy est.
B_{2025}	3,100	3,000	3,000
$CV_{B_{2025}}$	0.13	0.13	0.13
B_{MSY}	2,310	2,584	2,290
$CV_{B_{MSY}}$	0.3	0.35	0.3
B_{2025}/B_{MSY}	135%	116%	131%
B_0	5,975	6,568	5,929
$B_{35\%}$	2,066	1,844	2,057
SPR rate at F_{MSY}	31%	35%	31%
Steepness	0.62	0.59	0.62
Est. $B_{2024}/B_{2024, no fishing}$	0.61	0.54	0.59
B_{2024}/B_{MSY}	147%	127%	141%

Alternative assessment software

Stock-synthesis 3

Stock synthesis 3 (SS3) is a well establish framework and is common in many diverse settings. (Methot and Wetzel (2013)) For EBS pollock, we extended an initial framework based on the configurations used for Pacific hake (Grandin et al. (2024)) since there are many similarities in the type of data available. Namely, empirical weight-at-age, acoustic survey data, and indices of 1-year olds.

Within the SS3 framework, we evaluated different configurations for fishery selectivity variability (Figure 21). These showed the impact on the uncertainty and stock trends (Figure 22 for SSB and Figure 23 for recruitment).

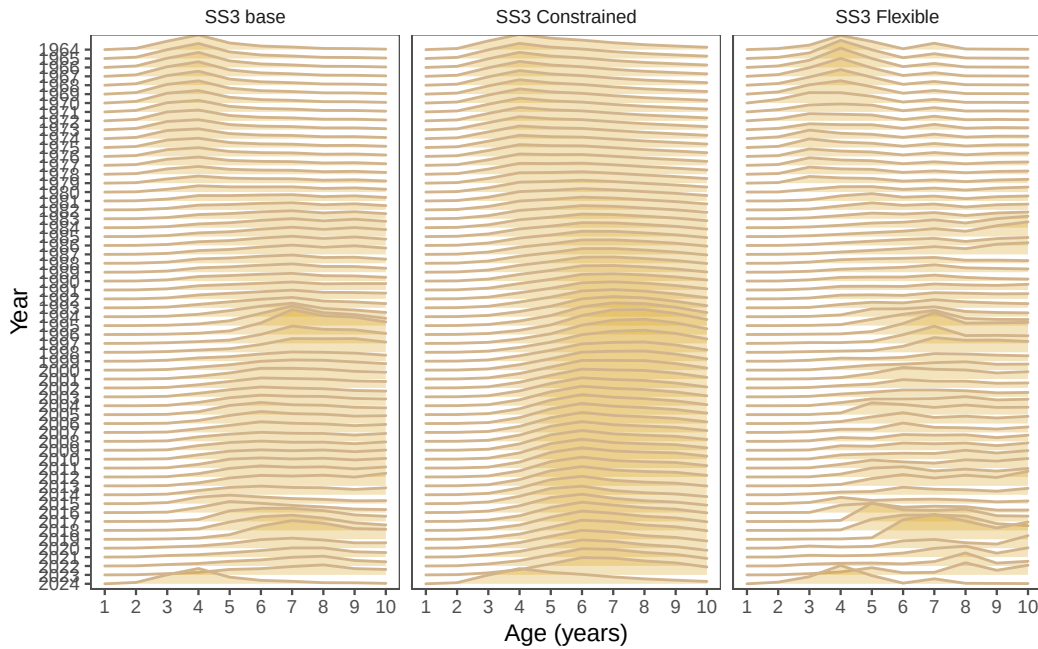


Figure 21: Model results comparing three different SS3 model configurations. The models are the base model (Base), the constrained model (Constrained), and the flexible model (Flexible).

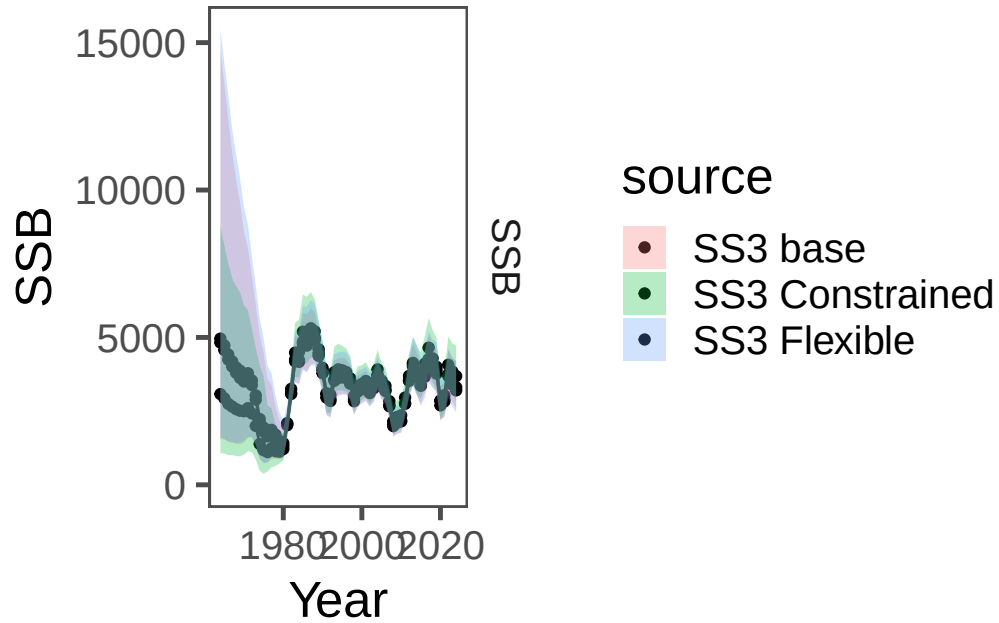


Figure 22: Model results comparing the time-series for three different SS3 model configurations. The models are the base model (Base), the constrained model (Constrained), and the flexible model (Flexible).

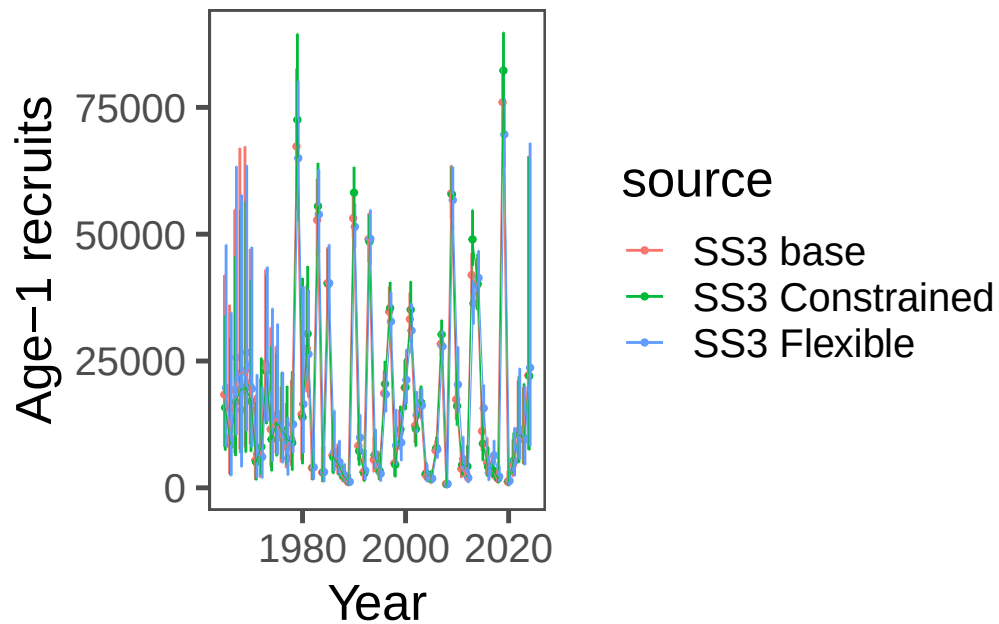


Figure 23: Model results comparing the time-series for three different SS3 model configurations. The models are the base model (Base), the constrained model (Constrained), and the flexible model (Flexible).

Comparing the “base” SS3 model with the pollock model showed different patterns in the early periods for selectivity (Figure 24) but slight differences in SSB and recruitment estimates (Figure 25 and Figure 26).

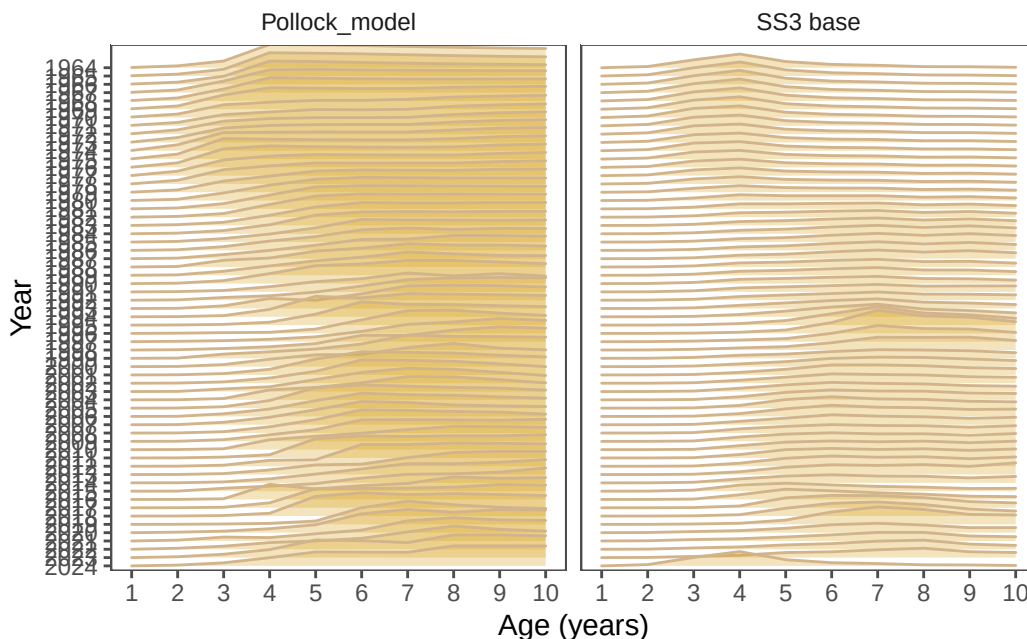


Figure 24: Model results comparing selectivity estimates between the base SS3 model configuration and the pollock model accepted in 2024.

SAM

The Stock Assessment Model (SAM) is a state-space model written in TMB (Nielsen and Berg (2014), Nielsen et al. (2021)). We attempted to configure the settings to have similar levels of flexibility to the reference model used for catch advice in 2024 and the settings and results can be found [here](#)). Comparing “selectivity” in the SAM model to the reference model, we see broad similarities but slightly lower inter-annual variability in the SAM run (Figure 27).

The spawning biomass trend largely overlaps in the uncertainty between the SAM and pollock model (Figure 28). The recruitment estimates are similar as well (Figure 29).

	type	source	se	value	Year
1	Biomass	CEA, FE	598.17756	3439.3743	1964
2	Biomass	CEA, FE	527.87773	4308.0457	1965
3	Biomass	CEA, FE	489.70388	4912.3713	1966
4	Biomass	CEA, FE	494.03978	6013.3280	1967

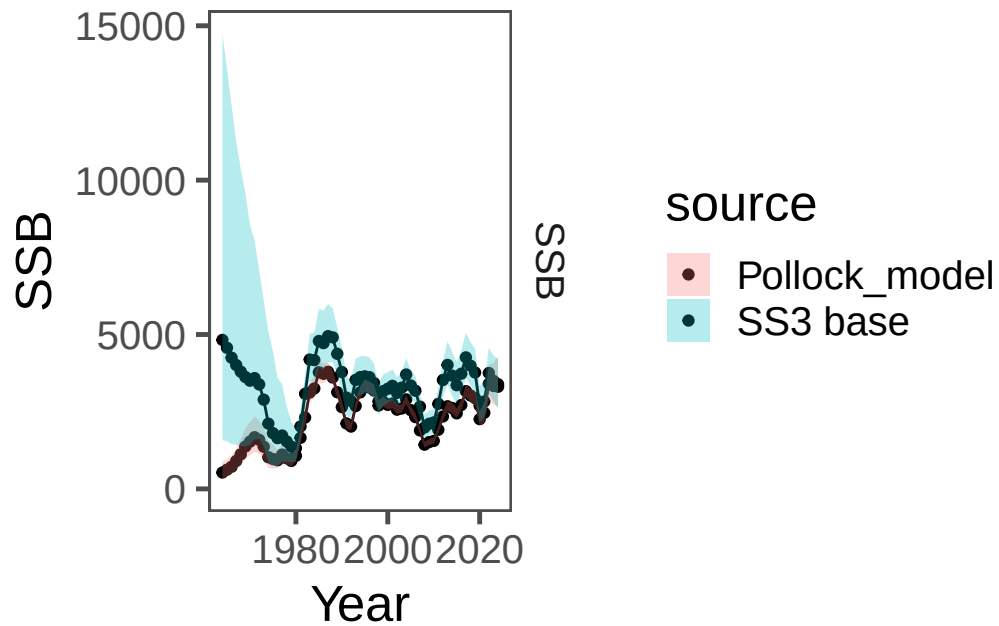


Figure 25: Model results comparing the base SS3 model configuration with the model used for pollock in 2024.

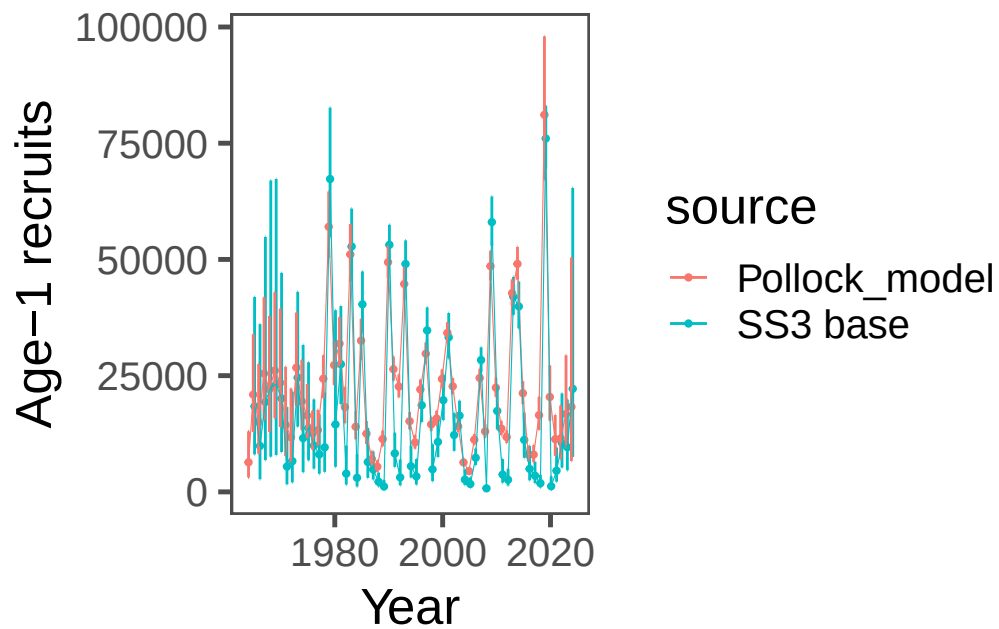


Figure 26: Model results comparing the base SS3 model configuration with the model used for pollock in 2024.

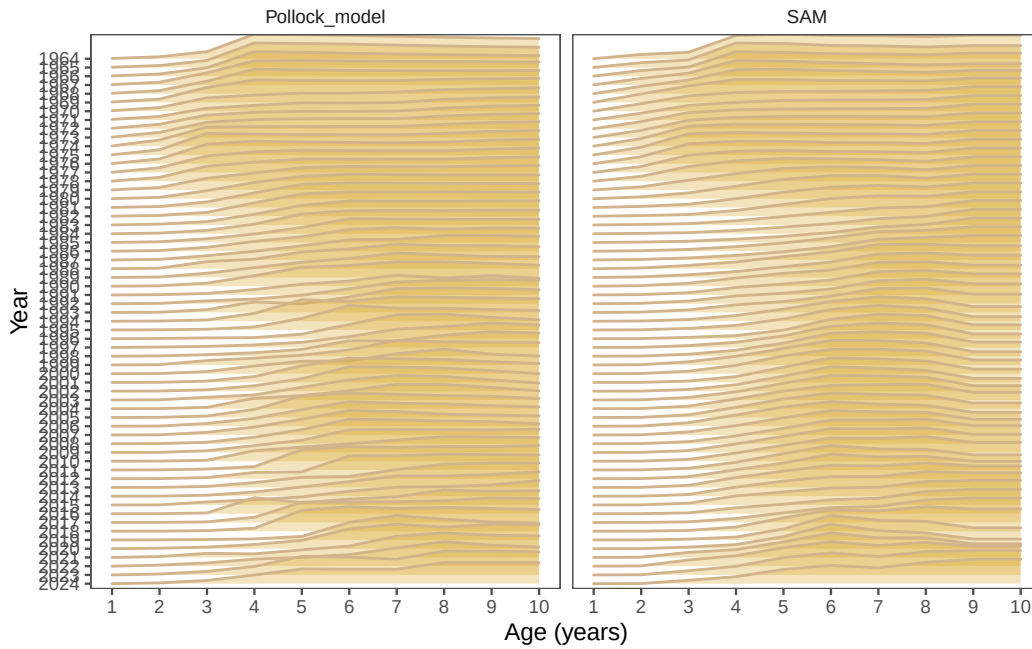


Figure 27: Model results comparing last year's model with the SAM model.

5	Biomass CEA, FE	532.92400	6969.1707	1968
6	Biomass CEA, FE	536.81933	7701.4589	1969
7	Biomass CEA, FE	557.82544	8534.3333	1970
8	Biomass CEA, FE	549.78156	8506.2570	1971
9	Biomass CEA, FE	525.71007	7769.6392	1972
10	Biomass CEA, FE	469.16874	6977.6100	1973
11	Biomass CEA, FE	419.40173	5905.0511	1974
12	Biomass CEA, FE	361.17764	5026.7801	1975
13	Biomass CEA, FE	296.02148	4475.7364	1976
14	Biomass CEA, FE	248.40359	4103.9810	1977
15	Biomass CEA, FE	229.75066	4303.2954	1978
16	Biomass CEA, FE	212.61136	5517.9693	1979
17	Biomass CEA, FE	204.64407	6322.2711	1980
18	Biomass CEA, FE	231.25071	7922.8261	1981
19	Biomass CEA, FE	284.58651	9567.1556	1982
20	Biomass CEA, FE	328.12244	11015.7151	1983
21	Biomass CEA, FE	330.20065	10806.5357	1984
22	Biomass CEA, FE	386.68764	12882.0868	1985
23	Biomass CEA, FE	362.30880	11770.4904	1986
24	Biomass CEA, FE	325.01055	11744.3525	1987
25	Biomass CEA, FE	298.67476	10734.6996	1988
26	Biomass CEA, FE	271.66492	9254.7113	1989

27	Biomass CEA, FE	249.10075	8901.5154	1990
28	Biomass CEA, FE	262.00036	9884.7445	1991
29	Biomass CEA, FE	267.66529	10430.5415	1992
30	Biomass CEA, FE	271.21434	10603.9665	1993
31	Biomass CEA, FE	267.77579	10405.8978	1994
32	Biomass CEA, FE	230.25118	8469.3684	1995
33	Biomass CEA, FE	235.26339	8549.1984	1996
34	Biomass CEA, FE	218.96539	7943.5994	1997
35	Biomass CEA, FE	221.05305	8155.2409	1998
36	Biomass CEA, FE	223.92060	8514.3632	1999
37	Biomass CEA, FE	232.45297	9191.0886	2000
38	Biomass CEA, FE	228.81074	9254.6976	2001
39	Biomass CEA, FE	234.93795	9913.1315	2002
40	Biomass CEA, FE	261.17480	11416.5248	2003
41	Biomass CEA, FE	248.90825	10603.4771	2004
42	Biomass CEA, FE	216.65365	8781.8837	2005
43	Biomass CEA, FE	198.33784	7481.9211	2006
44	Biomass CEA, FE	194.07004	7033.9708	2007
45	Biomass CEA, FE	166.34360	5747.1408	2008
46	Biomass CEA, FE	195.92949	6981.7691	2009
47	Biomass CEA, FE	239.21780	9168.7340	2010
48	Biomass CEA, FE	257.68739	10059.8552	2011
49	Biomass CEA, FE	272.89309	10253.5790	2012
50	Biomass CEA, FE	302.18755	11181.0921	2013
51	Biomass CEA, FE	311.24082	12050.4492	2014
52	Biomass CEA, FE	324.71242	13024.1317	2015
53	Biomass CEA, FE	341.68890	13764.8013	2016
54	Biomass CEA, FE	324.71644	12420.6702	2017
55	Biomass CEA, FE	296.03558	10468.6550	2018
56	Biomass CEA, FE	300.44402	10687.9255	2019
57	Biomass CEA, FE	316.67426	10874.2267	2020
58	Biomass CEA, FE	334.30851	10410.3203	2021
59	Biomass CEA, FE	408.94727	11265.9671	2022
60	Biomass CEA, FE	421.92922	10146.7683	2023
61	Biomass CEA, FE	543.08471	9765.4338	2024
62	SSB CEA, FE	190.87167	863.5953	1964
63	SSB CEA, FE	167.78091	1070.9124	1965
64	SSB CEA, FE	168.79696	1247.6196	1966
65	SSB CEA, FE	170.36528	1482.5471	1967
66	SSB CEA, FE	192.62868	1746.4223	1968
67	SSB CEA, FE	210.26197	2010.8458	1969
68	SSB CEA, FE	208.87197	2122.3618	1970
69	SSB CEA, FE	217.68088	2250.4911	1971

70	SSB CEA, FE	201.71375	2139.8496	1972
71	SSB CEA, FE	192.23652	1916.2478	1973
72	SSB CEA, FE	177.27196	1538.3620	1974
73	SSB CEA, FE	167.26824	1408.1008	1975
74	SSB CEA, FE	126.52876	1216.5327	1976
75	SSB CEA, FE	133.80915	1338.4520	1977
76	SSB CEA, FE	112.85780	1143.4891	1978
77	SSB CEA, FE	93.93469	1064.0355	1979
78	SSB CEA, FE	85.76932	1282.6114	1980
79	SSB CEA, FE	80.63457	1947.5565	1981
80	SSB CEA, FE	94.77074	2703.1509	1982
81	SSB CEA, FE	127.72116	3662.0451	1983
82	SSB CEA, FE	139.32211	3812.6062	1984
83	SSB CEA, FE	155.85742	4363.4659	1985
84	SSB CEA, FE	152.69610	4284.8762	1986
85	SSB CEA, FE	147.37058	4417.1832	1987
86	SSB CEA, FE	133.56934	4288.8246	1988
87	SSB CEA, FE	120.34325	3822.2585	1989
88	SSB CEA, FE	111.19063	3339.9248	1990
89	SSB CEA, FE	101.34997	2766.0956	1991
90	SSB CEA, FE	88.01350	2555.1920	1992
91	SSB CEA, FE	94.71360	3114.1209	1993
92	SSB CEA, FE	100.80089	3387.5588	1994
93	SSB CEA, FE	100.86929	3380.1055	1995
94	SSB CEA, FE	96.66224	3146.3550	1996
95	SSB CEA, FE	94.37793	2968.3560	1997
96	SSB CEA, FE	81.74393	2535.9808	1998
97	SSB CEA, FE	82.84627	2762.2338	1999
98	SSB CEA, FE	80.04559	2788.5809	2000
99	SSB CEA, FE	84.01575	2954.8660	2001
100	SSB CEA, FE	81.34538	2844.1541	2002
101	SSB CEA, FE	80.65924	2932.4907	2003
102	SSB CEA, FE	86.58040	3245.6868	2004
103	SSB CEA, FE	78.78259	2860.9129	2005
104	SSB CEA, FE	77.17262	2646.0761	2006
105	SSB CEA, FE	72.05430	2231.1689	2007
106	SSB CEA, FE	63.54837	1779.2715	2008
107	SSB CEA, FE	67.05085	1940.4768	2009
108	SSB CEA, FE	67.24250	2005.1880	2010
109	SSB CEA, FE	76.32899	2444.2387	2011
110	SSB CEA, FE	89.42754	2960.5153	2012
111	SSB CEA, FE	103.90316	3384.9907	2013
112	SSB CEA, FE	108.29424	3364.8423	2014

113	SSB CEA, FE	99.67968	3122.6491	2015
114	SSB CEA, FE	101.38400	3319.2674	2016
115	SSB CEA, FE	111.89005	3690.6429	2017
116	SSB CEA, FE	110.72711	3426.9770	2018
117	SSB CEA, FE	114.82409	3255.5603	2019
118	SSB CEA, FE	96.66492	2434.2877	2020
119	SSB CEA, FE	92.21926	2468.2623	2021
120	SSB CEA, FE	122.26997	3248.7539	2022
121	SSB CEA, FE	133.16426	3161.3806	2023
122	SSB CEA, FE	163.67251	3208.3073	2024
123	Recruits CEA, FE	3111.66347	6694.6821	1964
124	Recruits CEA, FE	6188.79694	27055.8182	1965
125	Recruits CEA, FE	5189.02841	16000.6137	1966
126	Recruits CEA, FE	6656.18057	26928.4899	1967
127	Recruits CEA, FE	6390.37696	23195.5687	1968
128	Recruits CEA, FE	6837.07603	27268.9306	1969
129	Recruits CEA, FE	6221.93235	23875.3560	1970
130	Recruits CEA, FE	4623.87084	14017.0964	1971
131	Recruits CEA, FE	3553.84684	10203.9050	1972
132	Recruits CEA, FE	4131.74442	21850.2979	1973
133	Recruits CEA, FE	2891.72832	14740.7878	1974
134	Recruits CEA, FE	2345.39781	13819.8700	1975
135	Recruits CEA, FE	2014.66085	12742.5522	1976
136	Recruits CEA, FE	1933.47535	14153.7468	1977
137	Recruits CEA, FE	2269.58880	25413.7786	1978
138	Recruits CEA, FE	3240.49173	61631.5070	1979
139	Recruits CEA, FE	2294.81720	30340.7577	1980
140	Recruits CEA, FE	2501.79434	36111.8063	1981
141	Recruits CEA, FE	1793.35744	18429.3440	1982
142	Recruits CEA, FE	3015.25402	54972.2840	1983
143	Recruits CEA, FE	1600.61054	16107.1135	1984
144	Recruits CEA, FE	2296.13318	40008.8483	1985
145	Recruits CEA, FE	1305.08445	15074.7616	1986
146	Recruits CEA, FE	853.80918	8559.7067	1987
147	Recruits CEA, FE	671.56176	6674.0336	1988
148	Recruits CEA, FE	866.97932	12380.1168	1989
149	Recruits CEA, FE	1626.03031	49835.3664	1990
150	Recruits CEA, FE	1106.57871	24019.0193	1991
151	Recruits CEA, FE	976.31035	19027.8505	1992
152	Recruits CEA, FE	1477.46135	39486.7150	1993
153	Recruits CEA, FE	834.55396	14520.0837	1994
154	Recruits CEA, FE	710.98814	11061.9600	1995
155	Recruits CEA, FE	1035.77701	23565.6881	1996

156	Recruits	CEA, FE	1224.52357	33309.8411	1997
157	Recruits	CEA, FE	824.11162	16331.7217	1998
158	Recruits	CEA, FE	841.03967	17633.9557	1999
159	Recruits	CEA, FE	1005.41332	25969.7543	2000
160	Recruits	CEA, FE	1133.77260	35364.9076	2001
161	Recruits	CEA, FE	869.70466	23649.8979	2002
162	Recruits	CEA, FE	661.98546	15523.5766	2003
163	Recruits	CEA, FE	418.22239	7097.6097	2004
164	Recruits	CEA, FE	359.40357	5185.5070	2005
165	Recruits	CEA, FE	629.09387	13055.5610	2006
166	Recruits	CEA, FE	1098.51578	29384.2252	2007
167	Recruits	CEA, FE	750.16343	13836.9683	2008
168	Recruits	CEA, FE	1823.59914	55238.2687	2009
169	Recruits	CEA, FE	1158.77034	25804.3838	2010
170	Recruits	CEA, FE	860.66650	15731.1723	2011
171	Recruits	CEA, FE	721.43589	12880.5225	2012
172	Recruits	CEA, FE	1477.07553	46372.1722	2013
173	Recruits	CEA, FE	1597.04792	49874.1413	2014
174	Recruits	CEA, FE	985.79624	20194.7800	2015
175	Recruits	CEA, FE	562.01442	7100.5741	2016
176	Recruits	CEA, FE	591.25944	6864.9109	2017
177	Recruits	CEA, FE	925.68818	12837.3220	2018
178	Recruits	CEA, FE	3215.10320	74603.0707	2019
179	Recruits	CEA, FE	1853.93433	20275.1711	2020
180	Recruits	CEA, FE	1849.65860	13255.9447	2021
181	Recruits	CEA, FE	2066.01367	10978.7076	2022
182	Recruits	CEA, FE	3576.00025	16263.4065	2023
183	Recruits	CEA, FE	14510.48069	51371.3228	2024

Other models in development

The “Woods Hole Assessment model” (WHAM) is written in TMB and is a fully state-space model (Stock and Miller (2021)). Initial trials are available on the [github repository](#).

The Rceattle model is developed by the AFSC and is written in TMB (Adams et al. (2022)). The model is an extension of the ADMB version used for multi-species model in the Eastern Bering Sea (Jurado-Molina, Livingston, and Ianelli (2005), Holsman et al. (2024)) and has a number of newer features. It is generalizable to include a variety of data types using lengths and can be set up for species tracked for sexual dimorphic growth and other processes. Since it has been recast into TMB, it allows for the ability to specify random effects in a number of processes. One development of the model is to have the flexibility to bridge among extant

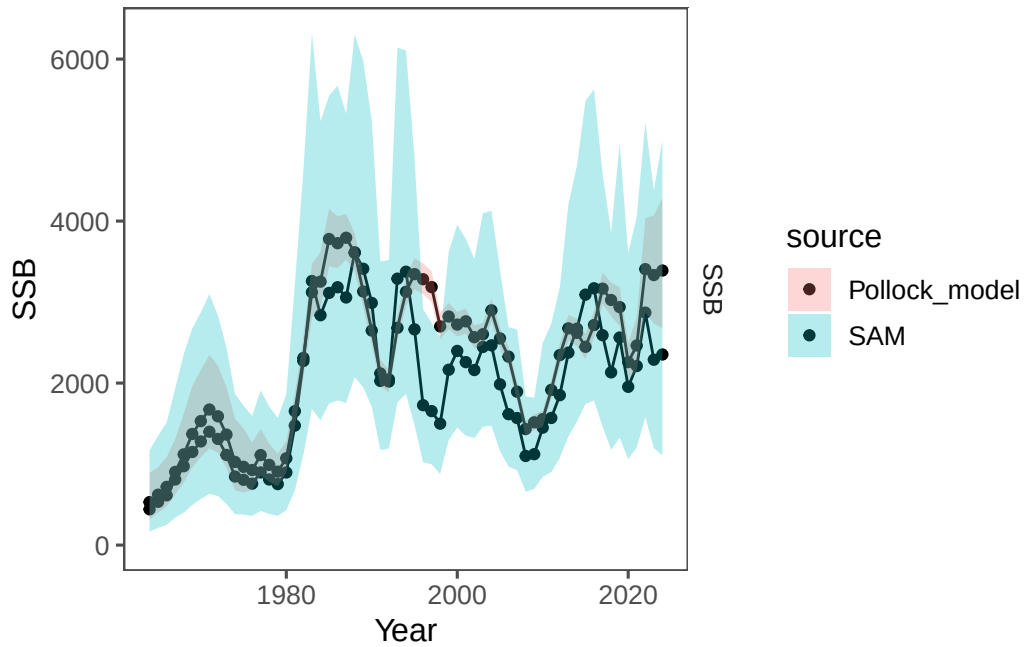


Figure 28: Model results comparing last year's model with the SAM model for spawning biomass (top) and recruitment at age 1 (bottom).

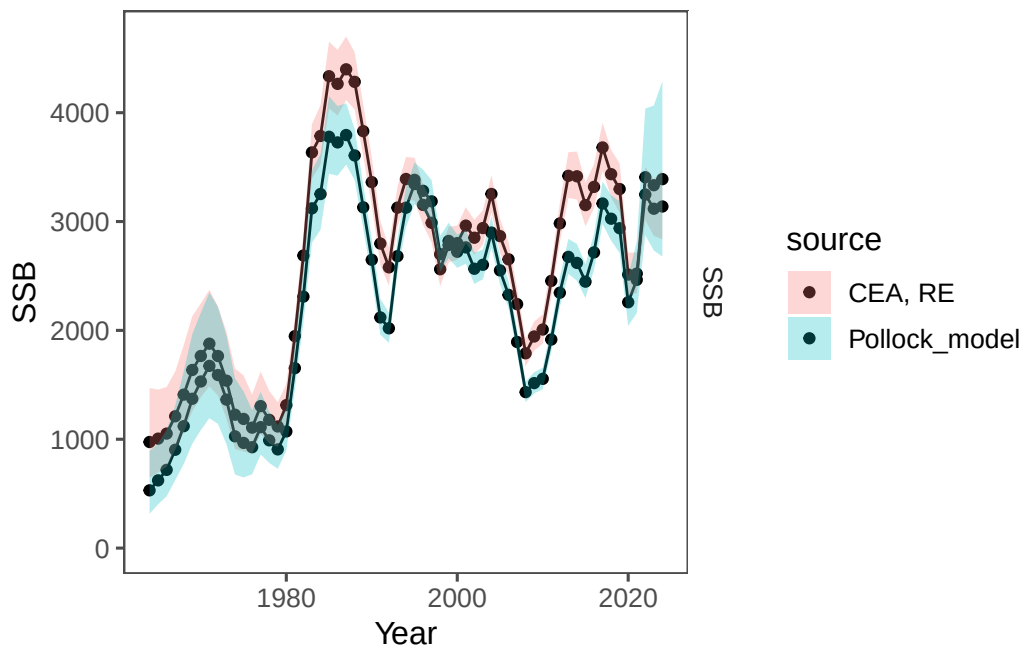


Figure 29: Model results comparing last year's model with the SAM model for spawning biomass (top) and recruitment at age 1 (bottom).

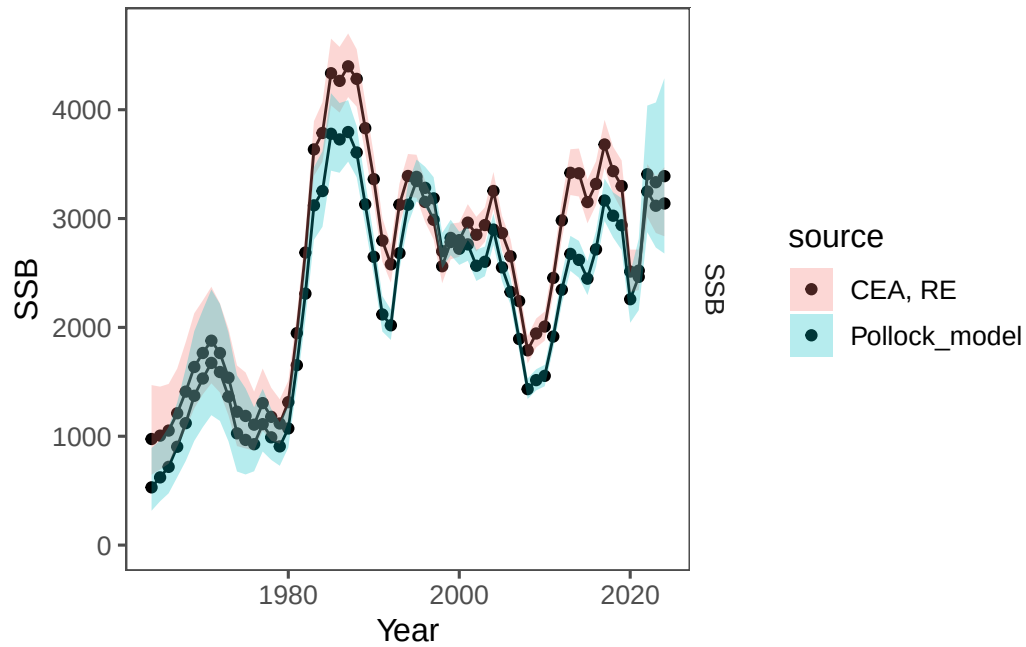


Figure 30: Model results comparing last year's model with the SAM model for spawning biomass (top) and recruitment at age 1 (bottom).

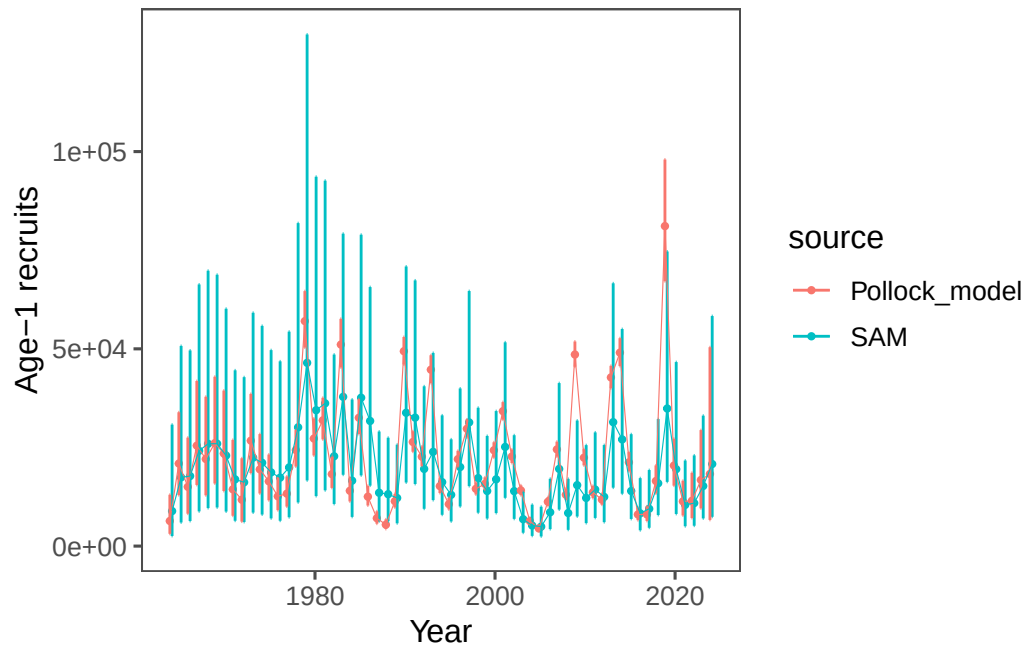


Figure 31: Model results comparing last year's model with the SAM model for spawning biomass (top) and recruitment at age 1 (bottom).

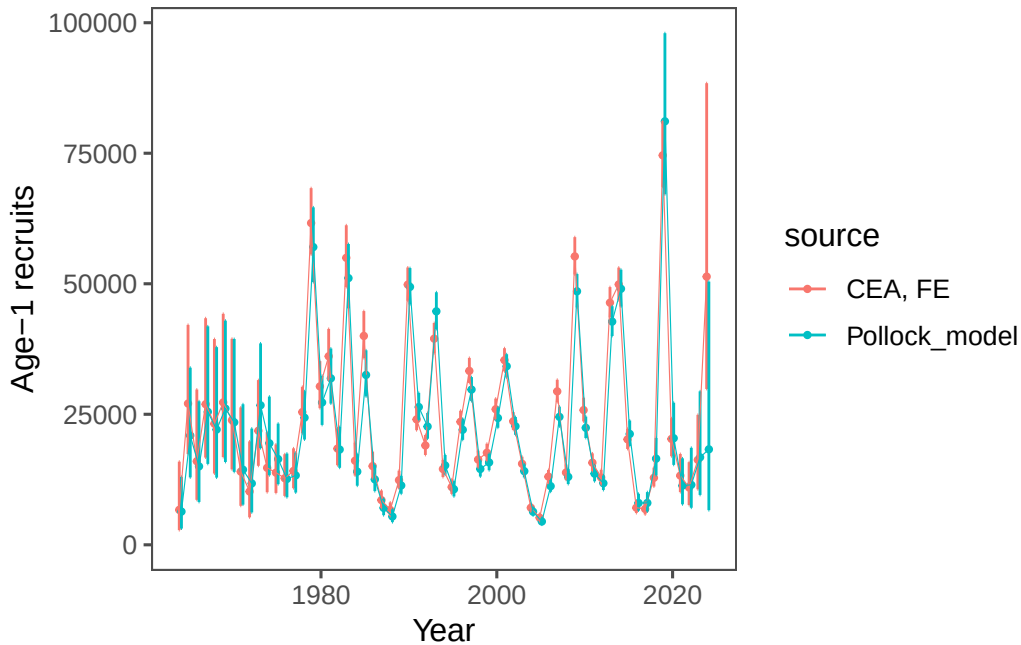


Figure 32: Model results comparing last year’s model with the SAM model for spawning biomass (top) and recruitment at age 1 (bottom).

single-species modeling platforms by using it in “single-species” mode. To illustrate that, we introduce an application of Rceattle to the EBS pollock data set. The model is not yet fully developed, but we present preliminary results for consideration. Initial runs are favorable but further work is needed to bridge the selectivity options.

Spatial modeling of the EBS pollock stock

In development is a spatially disaggregated general model by colleagues at the Auke Bay lab and the University of Alaska, Fairbanks (M. Cheng (2025)). This general model which has flexibility to specify regions and a variety of options including random effects. While this model is not yet fully developed, we present preliminary results here. The model is written in RTMB and is similar to the WHAM model. We pursue this to explore the potential for a more flexible model that can be used for more explicit considerations of stocks linked to different regions (e.g., the US and Russian portions of the EBS). Also, this model to add sex-specificity and random effects (M. L. H. Cheng et al. (2025)).

Figure 36 shows the model results comparing time-varying selectivity whereas Figure 37 shows the model results time series. The difference in the age-1 recruits (Figure 38) is likely due to the fact that the spatial model currently has constant M-at-age compared to the base model.

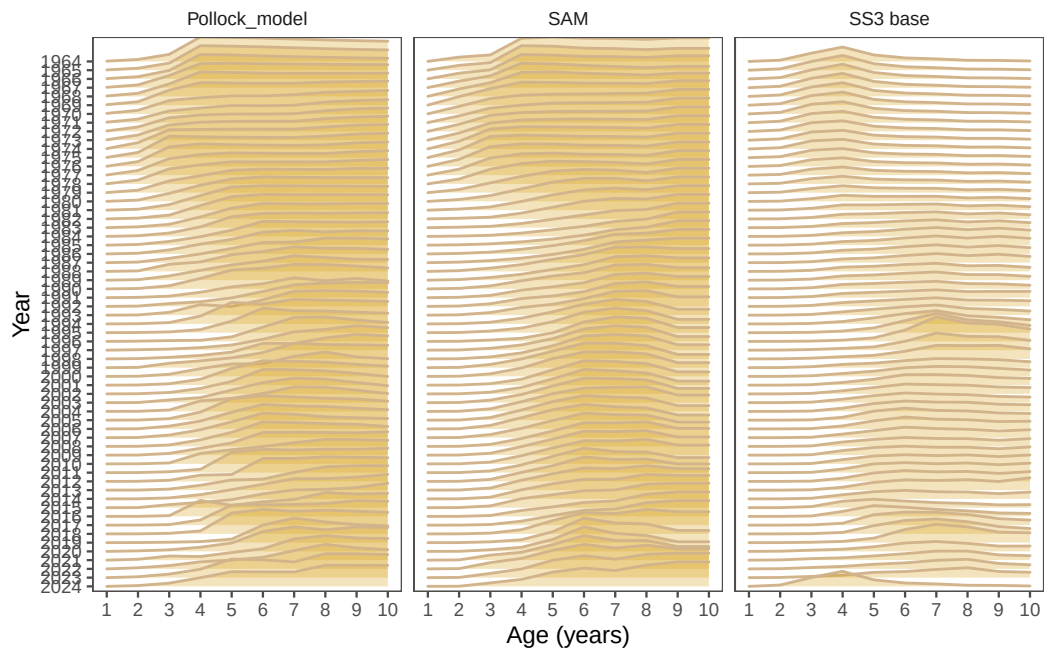


Figure 33: Model results comparing last year's model with the Rceattle model.

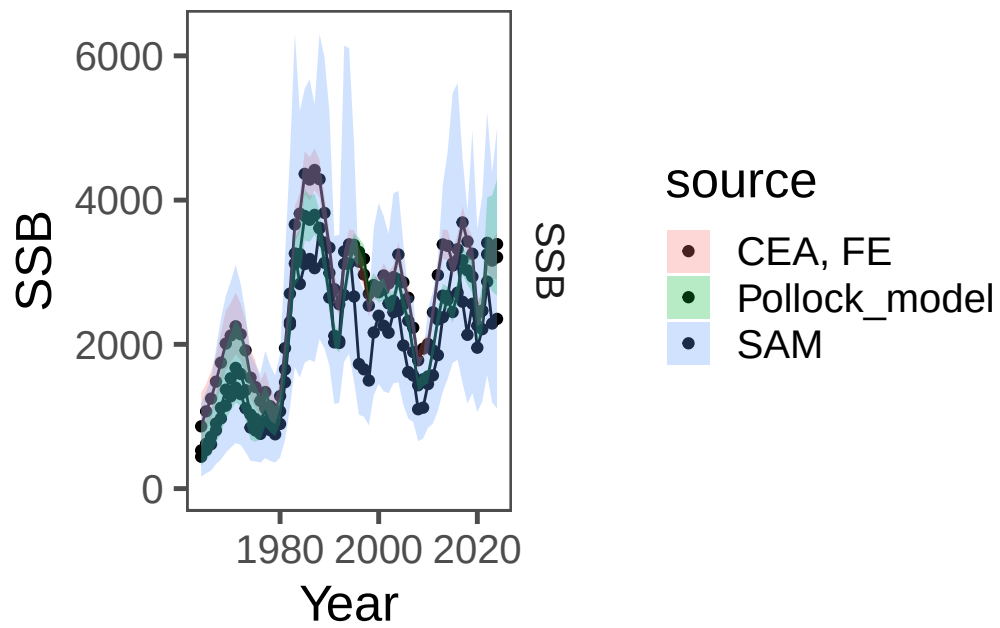


Figure 34: Model results comparing last year's model with the Rceattle model.

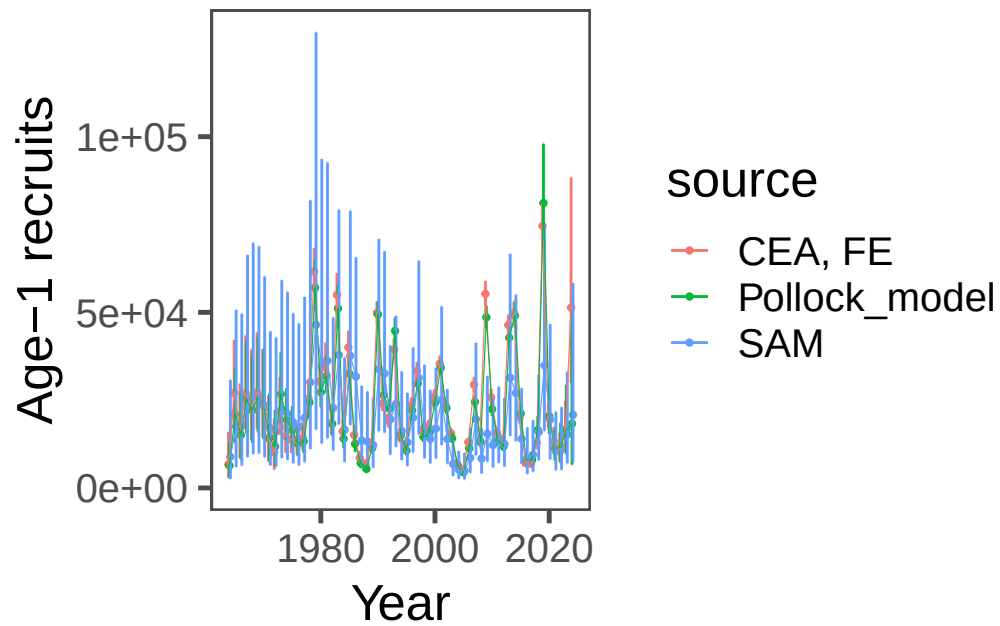


Figure 35: Model results comparing last year's model with the Rceattle model.

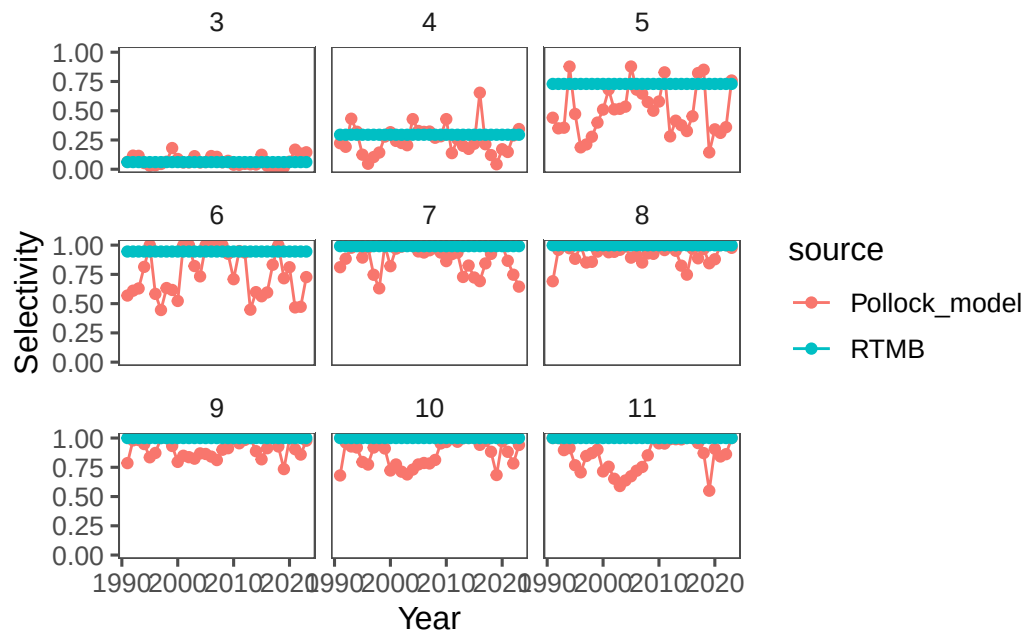


Figure 36: Model results comparing last year's model with the SPock model.

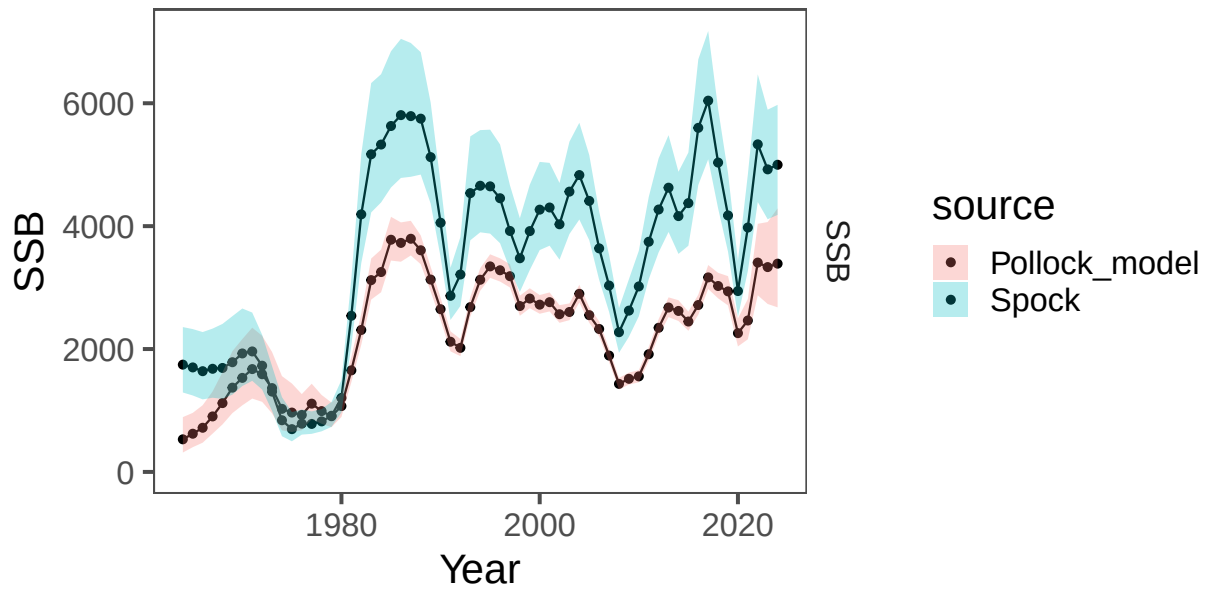


Figure 37: Model results comparing last year's model with the SPock model.

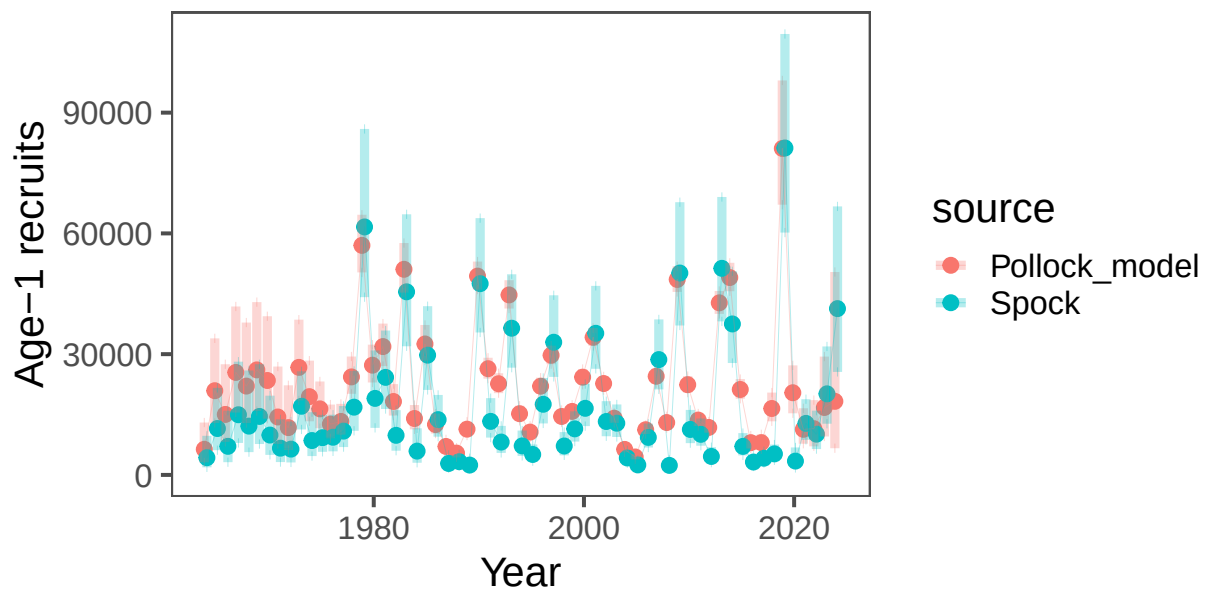


Figure 38: Model results comparing last year's model with the SPock model.

Bottom-trawl survey spatial distribution modeling

Abundance Index Methods

For model-based indices in the Bering Sea, we fitted observations of numerical abundance or biomass per unit area (where the choice of abundance or biomass varied by stock at the request of assessors) from all sampling locations in the 83-112 bottom trawl survey of the EBS, 1982-2025, including:

- Exploratory northern extension samples in 2001, 2005, and 2006
- 83-112 samples available in the NBS in 1982, 1985, 1988, 1991, 2010, 2017-2019, 2021-2023, and 2025

NBS samples collected prior to 2010 and in 2018 did not follow the 20 nautical mile sampling grid that was used in other survey years. Assimilating these unbalanced data requires extrapolating into unsampled areas, facilitated by including a spatially varying response to cold-pool extent (Thorson 2019). This spatially varying coefficient term was estimated for both linear predictors of a delta-model, and detailed comparison of results for EBS pollock has shown that it has a small but notable effect on these indices and resulting stock assessment outputs (O’Leary et al. 2020).

Rather than using the cold-pool covariate for yellowfin sole, we instead used the mean bottom temperature within the inner and middle domain strata from an interpolated temperature product. All models were fitted in the sdmTMB R package (<https://pbs-assess.github.io/sdmTMB>; Anderson et al. (2022)). All environmental data used as covariates were computed within the coldpool R package (<https://github.com/afsc-gap-products/coldpool>; Rohan, Barnett, and Charriere (2022)).

We used a spatiotemporal Poisson-link delta-model (Thorson 2018) involving two linear predictors and a gamma distribution to model positive catch rates. We predicted population density across the entire EBS and NBS in each year, using AFSC GAP-approved prediction grids. Spatial and spatiotemporal fields were estimated using a spatial mesh including 250 “knots” whose locations were determined using a K-means algorithm to place the knots evenly over space, in proportion to the dimensions of the prediction grid.

While the prior VAST model was fitted with a 750 knot mesh, we found that sdmTMB models fitted with mesh resolutions from 250-750 knots all provided extremely similar index estimates and that reducing the number of knots dramatically improved computation time. We estimated geometric anisotropy (how spatial autocorrelation declines with differing rates over distance in some cardinal directions than others) and included a spatial and spatiotemporal term for both linear predictors.

To facilitate prediction of density in regions with unsampled years, we specified that the spatiotemporal fields were structured over time as an AR(1) process (where the magnitude of autocorrelation was estimated as a fixed effect for each linear predictor). However, in cases where the AR(1) correlation parameter was estimated to be close to 1 for either model component, the model would be collapsed into a simpler structure by specifying $\rho = 1$, i.e., modeling spatiotemporal variation as a random walk. We did not include any temporal correlation for intercepts, which we treated as fixed effects for each linear predictor and year. Finally, we used epsilon bias-correction to correct for retransformation bias (Thorson and Kristensen 2016).

We checked model fits for convergence by confirming that the derivative of the marginal likelihood with respect to each fixed effect was sufficiently small (less than ~ 0.001) and that the Hessian matrix was positive definite, along with other model checks produced by the function `sdmTMB::sanity()`. We then checked goodness of model fit by computing Dunn-Smyth randomized quantile residuals (Dunn and Smyth 1996) and visualizing these using a quantile-quantile plot within the DHARMA R package (Hartig 2021). We also evaluated the distribution of these residuals over space in each year, and inspected them for evidence of residual spatiotemporal patterns. Results compared favorably among methods, and the model index result was very similar to previous estimates (Figure 39).

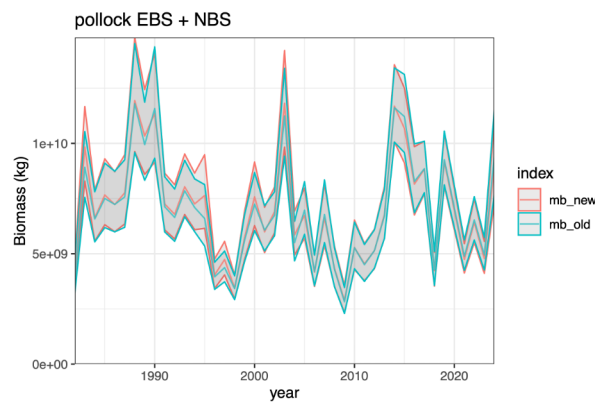


Figure 39: Model results comparing last year’s model with the revised sdmTMB model.

Model-based Age Composition Methods

For model-based estimation of age compositions in the Bering Sea, we fitted observations of numerical abundance-at-age at each sampling location. This was made possible by applying a year-specific, region-specific (EBS and NBS) age-length key to records of numerical abundance and length composition. In subcategories (combinations of year, length, age, sex) that contained insufficient data, age composition was computed from length composition given a globally pooled age-length key.

We computed these estimates in the `tinyVAST` R package (<https://vast-lib.github.io/tinyVAST>; Thorson et al. (2024)), assuming a Poisson-link delta-model (Thorson 2018) involving two linear predictors, and a gamma distribution to model positive catch rates. We did not include any density covariates in estimation of age composition for consistency with models used in the previous assessments and due to computational limitations.

We estimated the spatial range assuming geometric isotropy (i.e, spatial autocorrelation declines with equal rates over distance in all cardinal directions). We used the same extrapolation grid as implemented for abundance indices, but here we modeled spatial and spatiotemporal fields with a mesh with a coarser spatial resolution than the index model, here using 50 “knots”. This reduction in the spatial resolution of the model, relative to that used for the abundance indices, was necessary due to the increased computational load of fitting multiple age categories.

The estimates of abundance at age were bias-corrected with the same method as the abundance index. We implemented the same diagnostics to check convergence and model fit as those used for the abundance index. Results compared favorably among methods (Figure 40).

Both the model-based bottom trawl index and age composition estimates were calculated using code in the AFSC GAP model-based indices GitHub repository (<https://github.com/afsc-gap-products/model-based-indices>). Design-based estimates of bottom trawl products were calculated using code in the `gapindex` R package (<https://github.com/afsc-gap-products/gapindex>).

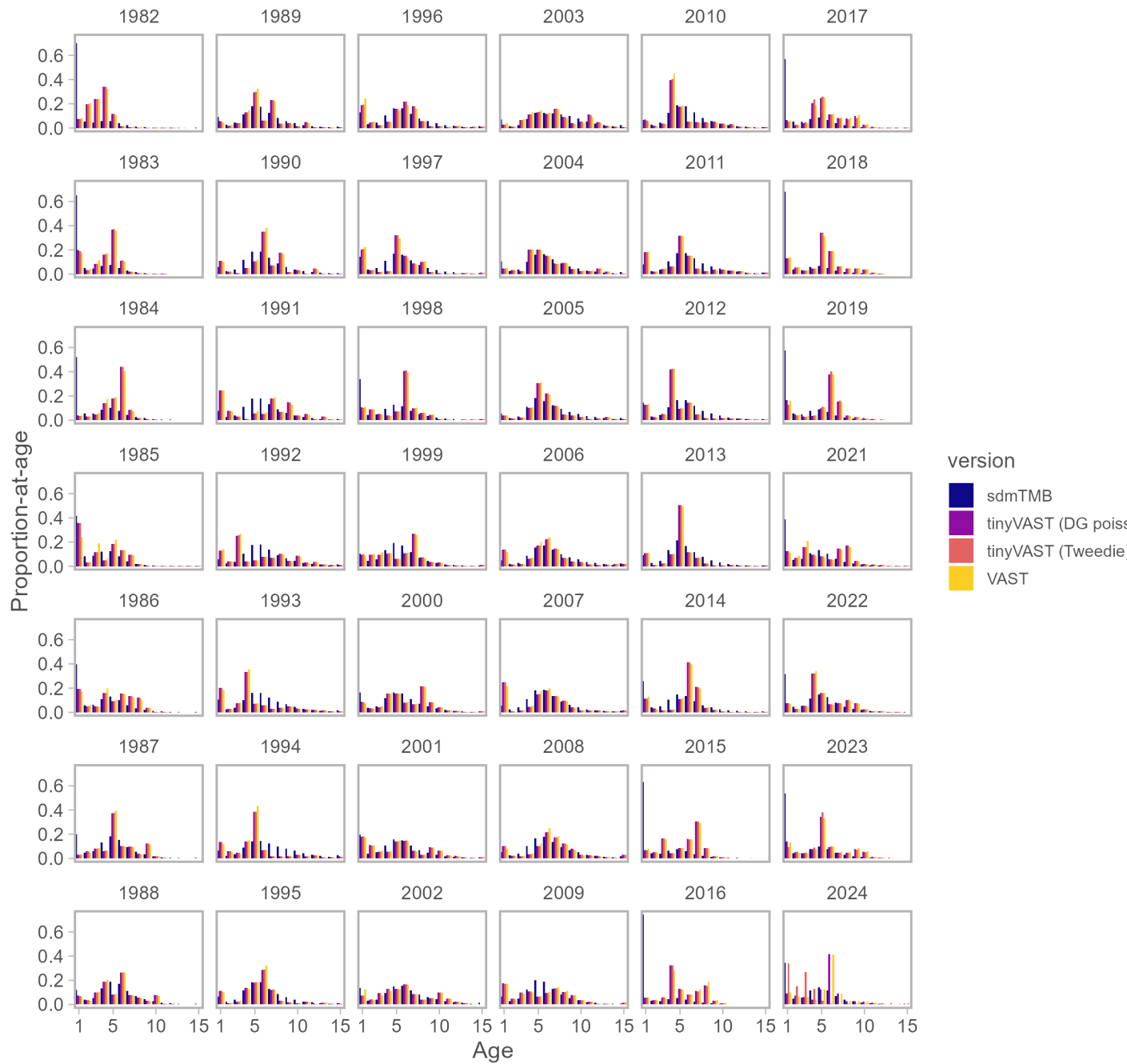


Figure 40: Model results comparing last year's model with the revised sdmTMB model.

Pollock model fishery selectivity notes

The following provides more detail based on the actual code as adopted from Butterworth, Ianelli, and Hilborn (2003).

Initial Setup

Average Selectivity

$$\overline{sel} = \log(\text{mean}(\exp(\mathbf{c})))$$

where $\mathbf{c}$ is the vector of base selectivity coefficients.

Base Year Selectivity ($t = \text{tsel}$)

For ages 1 to n_{sel} :

$$\log(S_{t,a}) = c_a \quad \text{for } a = 1, 2, \dots, n_{sel}$$

For older ages (plateau assumption):

$$\log(S_{t,a}) = c_{n_{sel}} \quad \text{for } a = n_{sel} + 1, \dots, n_{ages}$$

Initial Normalization

$$\log(S_{t,a}) = \log(S_{t,a}) - \log\left(\frac{1}{n_{ages}} \sum_{a=1}^{n_{ages}} \exp(\log(S_{t,a}))\right)$$

This ensures: $\prod_{a=1}^{n_{ages}} S_{t,a}^{1/n_{ages}} = 1$

Time-Varying Component

Selectivity Evolution ($t > \text{tsel}$)

For years without regime changes:

$$\log(S_{t+1,a}) = \log(S_{t,a}) \quad \text{for all ages}$$

For years with regime changes (when $t + 1 \in \text{yrs_ch_fsh}$):

Ages 1 to nsel:

$$\log(S_{t+1,a}) = \log(S_{t,a}) + \delta_{ii,a} \quad \text{for } a = 1, 2, \dots, \text{nsel}$$

Older ages (plateau maintained):

$$\log(S_{t+1,a}) = \log(S_{t+1,\text{nsel}}) \quad \text{for } a = \text{nsel} + 1, \dots, \text{nages}$$

where: - $\delta_{ii,a}$ is the selectivity deviation for change event ii and age a - ii is the index of the current selectivity change event

Annual Normalization

For each year t :

$$\log(S_{t,a}) = \log(S_{t,a}) - \log\left(\frac{1}{\text{nages}} \sum_{a=1}^{\text{nages}} \exp(\log(S_{t,a}))\right)$$

and the actual selectivity is:

$$S_{t,a} = \exp(\log(S_{t,a}))$$

Constraints and Properties

1. Normalization Constraint:

$$\prod_{a=1}^{\text{nages}} S_{t,a}^{1/\text{nages}} = 1 \quad \forall t$$

2. Dome-shaped Constraint:

$$S_{t,a} = S_{t,\text{nsel}} \quad \text{for } a > \text{nsel}$$

3. **Continuity:** Selectivity changes only occur in specified years (`yrs_ch_fsh`)
4. **Deviation Structure:**

$\delta_{ii} \sim$ estimated deviations with penalties

Implementation Notes

- Base coefficients $\mathbf{c}$ are estimated parameters
- Deviations δ are typically penalized to prevent overfitting
- The plateau assumption prevents unrealistic selectivity patterns for older ages
- Geometric mean normalization maintains interpretability relative to fishing mortality

Regularization and Penalty Terms

The selectivity model includes several penalty terms to ensure realistic and stable estimates:

Penalty Term 1: Deviation Magnitude Penalty $P_1 = \frac{\lambda_1}{N_{groups}} \sum_{i=1}^{N_{changes}} \sum_{a=1}^{n_{sel}} \delta_{i,a}^2$ where:

$\lambda_1 = \text{ctrl_flag}(10)$ (penalty weight parameter) $N_{groups} = \text{group_num_fsh}$ (number of fishery groups) This shrinks deviations toward zero when not informed by data

Penalty Term 2: Base Selectivity Smoothness Penalty $P_2 = \frac{\lambda_2}{N_{changes}} \sum_{a=1}^{n_{sel}-2} [\nabla^2 \log(S_{styr,a})]^2$ where:

$\lambda_2 = \text{ctrl_flag}(11)$ (smoothness penalty weight) $\nabla^2 \log(S_{t,a}) = \log(S_{t,a+2}) - 2\log(S_{t,a+1}) + \log(S_{t,a})$ (second difference) This penalizes high curvature in the base selectivity pattern

Penalty Term 3: Change Year Smoothness Penalty $P_3 = \frac{\lambda_2}{N_{changes}} \sum_{i=1}^{N_{changes}} \sum_{a=1}^{n_{sel}-2} [\nabla^2 \log(S_{t_i,a})]^2$

where t_i represents each change year. This maintains smooth selectivity curves in all change years. Penalty Term 4: Temporal Stability Penalty (Random Walk)

$P_4 = \sum_{i=1}^{N_{changes}} \frac{1}{2\sigma_i^2} \sum_{a=1}^{n_{ages}} [\log(S_{t_{i-1},a}) - \log(S_{t_i,a})]^2$ where:

$\sigma_i = \text{sel_ch_sig_fsh}(i)$ (change magnitude parameter for change event i) This penalizes large year-to-year changes in selectivity Only applied when $t_i \neq \text{styr}$ (change year is not the start year)

Total Penalty Function $P_{total} = P_1 + P_2 + P_3 + P_4$ The total negative log-likelihood becomes: $-\log L = -\log L_{data} + P_{total}$ Penalty Interpretation

P_1 : Prevents overfitting by shrinking deviations toward zero P_2 and P_3 : Enforce smooth, realistic selectivity curves (penalize jagged patterns) P_4 : Constrains temporal evolution (selectivity changes smoothly over time)

The penalty weights (λ_1, λ_2) and variance parameters (σ_i) control the balance between fitting data and maintaining realistic selectivity patterns. Implementation Notes

Base coefficients $\mathbf{c}$ are estimated parameters Deviations δ are penalized to prevent overfitting The plateau assumption prevents unrealistic selectivity patterns for older ages Geometric mean normalization maintains interpretability relative to fishing mortality Second differences approximate curvature (second derivatives) for smoothness penalties

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